

# **ZMD31050**

**Advanced Differential Sensor Signal Conditioner**

Functional Description

Preliminary

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**Rev. 0.82, Preliminary**

This Revision is not valid for earlier revisions than ZMD31050C\*.  
For parts identification refer ZMD31050 Development Status Report.

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## 1. CONTROL LOGIC

### 1.1 General Description

The control logic of the ZMD31050 consists of the Calibration Micro-Controller (CMC) and the modules control logic of the A/D-Converter, Control logic of the digital Ports, PWM and serial digital Interface. The configuration of the various modes of the device is done by programming an EEPROM.

The Calibration Micro-Controller controls the measurement cycle and realizes the calculations for sensor signal conditioning. This eliminates the gain deviation, the offset, the temperature deviation and the non-linearity of the pre-amplified and A/D-converted sensor signal.

The communication of the ZMD31050 with an external micro-controller especially for calibration purposes is done via a serial digital interface. Communication protocols according to I<sup>2</sup>C and SPI standard are supported. Additionally an one wire interface is implemented. These serial interfaces are used for the calibration of the sensor system consisting of transducer and ZMD31050.

Furthermore the serial interface makes available the read out of the results of sensor signal conditioning as digital values during the calibration or during the Normal Operation Mode for processing by an external micro-controller. The internal processing of received interface commands is done by the CMC. As a consequence the measurement cycle is interrupted if a command is received. Only the read out of data is controlled by the serial interface itself and does not interrupt the CMC.

The controller of the A/D-conversion is started by the CMC. It generates a clock for the A/D-converter (switched capacitor technique), the chopper clock for the analog front end and executes a continuous measurement cycle. The result of the A/D-conversion is a counter value, which is read and processed after a synchronization signal which is generated after every measurement. The processing of the conditioning calculation by the CMC works parallel to the A/D-conversion.

The output of the conditioning result can be done via various channels: analog voltage output, (4 to 20) mA current loop, PWM. Additionally there are two digital output ports with programmable threshold, hysteresis and delay.

### 1.2 CMC Description

The Calibration Micro-Controller (CMC) is especially adapted to the tasks connecting with the signal conditioning. The main features are:

- Processing width 16 Bit
- Programmed via ROM
- Constants/coefficients for the conditioning calculation are stored in the EEPROM. After power-on or after re-initialization from EEPROM by sending a certain command to the serial interface the EEPROM is mirrored to the RAM.
- Continuous parity-check during every read from RAM. If incorrect data is detected, an error mode is started (error code is written to the serial digital output, the analog out is set to diagnostic mode).
- The conditioning calculation is done with 16 Bit saturation arithmetic, i.e. if an overflow occurs in any calculation the result is fixed to the maximum of the defined 16Bit range. The sign is preserved.

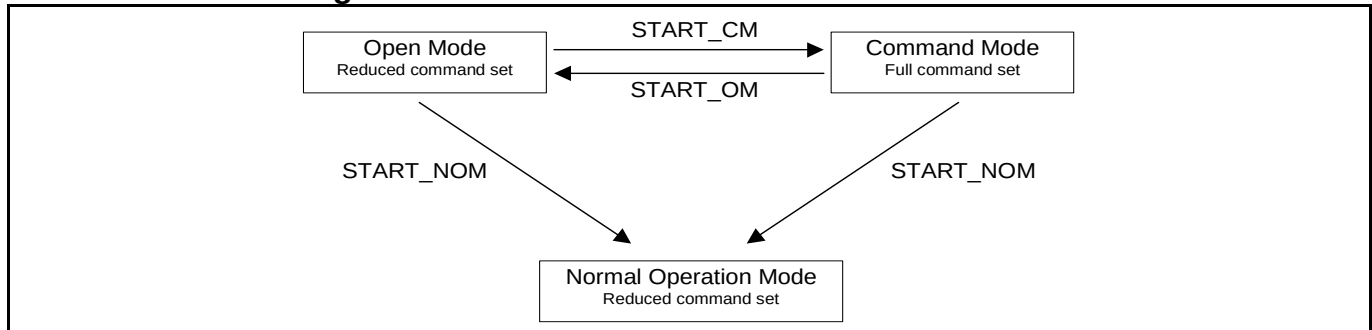
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### 1.3 General Working Modes



**Fig. 1:** Modes of digital serial interface

#### 1.3.1 Normal Operation Mode

The “Normal Operation Mode” (NOM) is the working mode in which the signal condition operation is done. After power-on the ZMD31050 starts with an initialization routine, which also can be started by sending a certain command via the serial digital interface at any time. First the EEPROM is mirrored to the RAM, thereby the EEPROM contents are checked. If an error is detected, the “Command Mode” is started, the analog output is shifted in the diagnose range and an error code can be read via the digital interface. Furthermore the configuration of the ZMD31050 is set during initialization. This configuration evaluates all the configuration data stored in the EEPROM, e.g. measurement cycle, signal output mode, mode of digital serial interface etc.. After the initialization the continuous measurement cycle, the conditioning calculation and the output via various channels starts. Provided that the EEPROM is programmed correct, this mode works without sending any command to the digital serial interface.

#### 1.3.2 Command Mode

The ZMD31050 changes to “Command Mode” (CM) after receiving a correct command via the digital serial interface. This command generates an interrupt for the CMC, the measurement cycle is stopped and the called command routine is processed. During processing the serial interface is disabled, no further commands are recognized. After finishing the routine the CMC waits for further commands. It is also possible to start the NOM by using the current contents of EEPROM or RAM.

#### 1.3.3 Digital Serial Interface

Regarding the digital serial interface 3 modes are distinguished: the “Open Mode” (OM), the “Normal Operation Mode” (NOM) and the “Command Mode” (CM). In OM and NOM only a reduced set of commands is available. For all commands which change the configuration of the ZMD31050 (write to EEPROM/RAM, start different measurement cycle) the ZMD31050 has to be previously transferred to CM. This protects the ZMD31050 against unintentional configurations.

After power-on the ZMD31050 starts in Open Mode. Every command sent to the digital serial interface transfers the ZMD31050 to the Normal Operation Mode except the command START\_CM, which starts the CM. The command START\_CM is only processed if it is received as first command after power-on via digital serial interface. Is the ZMD31050 once changed to NOM, this mode can only be reset by a new power-on. The ZMD31050 leaves the CM by receiving the commands START\_OM or START\_NOM or by reset due to power-on.

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If the ZMD31050 receives a command in NOM, which is not element of the reduced command set, this command is ignored, no interrupt for the CMC is generated and the measurement cycle is continued. If the serial interface works in I<sup>2</sup>C mode no acknowledges are generated after the command byte.

### 1.3.4 Error Codes

The ZMD31050 detects various possible errors. A detected error is signaled by changing in a diagnostic mode. In this case the analog output is set to High or Low (maximum or minimum possible output value) and the output registers of the digital serial interface are set to a significant error code (see Table 1). Note that the error detection functionality has to be partly enabled by configuration words (Sensor Connection; Common Mode Voltage out of limits).

| Detectable Error                  | Description                                                                                                | Enabled by [Register:Bit] | Sets SIF-Out to                                     | Sets Analog Out to <...> for Voltage Out Current Loop |       |
|-----------------------------------|------------------------------------------------------------------------------------------------------------|---------------------------|-----------------------------------------------------|-------------------------------------------------------|-------|
| EEPROM Sign                       | Signature-Check during read out of EEPROM after Power On or after SIF-Command COPY_EEP2RAM                 |                           | 0xCAAA                                              | ~VSS                                                  | >20mA |
| RAM Parity                        | Parity-Check at every RAM access                                                                           | CFGAPP: SCCD              | 0xCF0F                                              | ~VSS                                                  | >20mA |
| Register Parity                   | Permanent Parity-Check of Configuration Registers                                                          | CFGAPP: SCCD              | 0xCE38                                              | ~VSS                                                  | >20mA |
| ROM Check                         | Signature-Check of ROM after Power-on                                                                      | CFGCYC: ROMCHE            | 0xCCCC                                              | ~VSS                                                  | >20mA |
| Arithmetic Check                  | Arithmetic-Check during measurement cycle                                                                  |                           | 0xC1C7                                              | ~VSS                                                  | >20mA |
| Watchdog                          | Watchdog-Check of start-routine and measurement-cycle                                                      |                           | 0xC555                                              | ~VSS                                                  | >20mA |
| Configuration                     | Programmed configuration is not allowed (Conflict regarding A/D-conversion-time and enabled functionality) |                           | 0xC333                                              | ~VSS                                                  | >20mA |
| Sensor Connection                 | Connection-Check of Sensor Bridge                                                                          | CFGAPP: SCCD              | 0xCFCF                                              | ~VDDA                                                 | <4mA  |
| Common Mode Voltage out of limits | Check if Bridge Common Mode Voltage complies the programmed limits                                         | CGFCYC: ECMV              | 2'b10    V <sub>IN_CM</sub> /2                      | ~VDDA                                                 | <4mA  |
|                                   |                                                                                                            |                           | V <sub>IN_CM</sub> ... Measured Common Mode Voltage |                                                       |       |

**Table 1:** Error Codes

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## 2. SIGNAL CONDITIONING

### 2.1 A/D-Conversion

During Signal Conditioning Mode the analog preconditioned sensor signal is continuously A/D-converted. The A/D-conversion is configurable regarding resolution  $r_{ADC}$  and the inherent range shift  $RS_{ADC}$  by the configuration words. Furthermore the one or two step conversion mode is selectable (ADC order). The two step conversion is faster, the one step conversion is more accurate because of larger integration time. The selected configuration for the ADC is equal for all measurements (measurand (e.g. pressure), temperature 1 and 2, common mode voltage), thus the conversion time is constant during the whole measurement cycle. Only the inherent range shift is selectable individually for every single signal. Additionally for all measured signals auto-zero measurements are done during the measurement cycle. The resulting digital raw values for measurand (e.g. pressure) and temperature are defined by these equations:

⇒ **Analog differential input voltage to A/D-conversion**

Pressure:

$$V_{ADC\_DIFF\_p} = a_{IN\_p} \cdot V_{IN\_DIFF\_p} + a_{IN\_AZS} \cdot V_{AZS}$$

Temperature:

$$V_{ADC\_DIFF\_T} = a_{IN\_T} \cdot V_{IN\_DIFF\_T}$$

Residual Offset:

$$V_{ADC\_OFF} = a_{IN\_OFF} \cdot V_{IN\_OFF}$$

⇒ **Digital raw A/D-conversion result**

$$Z_{ADC} = 2^{r_{ADC}} \cdot \left( \frac{V_{ADC\_DIFF} + V_{ADC\_OFF}}{V_{ADC\_REF}} - 1 + RS_{ADC} \right)$$

⇒ **Auto-zero value**

$$Z_{AZ} = 2^{r_{ADC}} \cdot \left( \frac{V_{ADC\_OFF}}{V_{ADC\_REF}} - 1 + RS_{ADC} \right)$$

⇒ **Auto-zero corrected raw A/D-result**

$$Z_{CORR} = Z_{ADC} - Z_{AZ} = 2^{r_{ADC}} \cdot \frac{V_{ADC\_DIFF}}{V_{ADC\_REF}}$$

|                 |                                                                                 |
|-----------------|---------------------------------------------------------------------------------|
| $V_{IN\_DIFF}$  | Analog differential input voltage to analog front end                           |
| $V_{IN\_OFF}$   | Residual Offset voltage of analog front end                                     |
| ...             |                                                                                 |
| $V_{ADC\_DIFF}$ | Analog differential input voltage to A/D-converter                              |
| $V_{ADC\_OFF}$  | Residual offset voltage of analog front end to A/D-converter                    |
| $V_{ADC\_REF}$  | ADC-Reference voltage (ratiometric reference for pressure measurement)          |
| $r_{ADC}$ ...   | Resolution of A/D-conversion                                                    |
| $RS_{ADC}$      | Range Shift of A/D-conversion<br>(=1/2, 3/4, 7/8, 15/16)                        |
| $a_{IN\_p}$ ... | Gain of analog front end for pressure signal                                    |
| $a_{IN\_T}$ ... | Gain of analog front end for temperature signal                                 |
| $a_{IN\_OFF}$   | Gain of analog front end for residual offset                                    |
| $a_{IN\_AZS}$   | Gain of analog front end for analog zero point shift voltage                    |
| $V_{AZS}$       | Analog zero point shift voltage<br>(depending on $AZS_{IN} = (-25 \dots +25)$ ) |

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### 2.2 Correction Formula for Measurand (e.g. pressure)

The digital raw value for pressure is further processed with the correction formula to remove offset and temperature dependency and to compensate non-linearity up to 3<sup>rd</sup> order. The signal conditioning equation is solved by the CMC and is defined as follows:

⇒ Range definition of inputs

$$Z_{CORR\_p} \in [-2^{r_{ADC}}; 2^{r_{ADC}})$$

$$Z_{CORR\_T1} \in [-2^{r_{ADC}-1}; 2^{r_{ADC}-1})$$

$$c_i \in [-2^{15}; 2^{15})$$

⇒ Conditioning Equations

$$Y = \frac{Z_{CORR\_p} + c_0 + 2^{-(r_{ADC}-1)} \cdot c_4 \cdot Z_{CORR\_T1} + 2^{-2(r_{ADC}-1)} \cdot c_5 \cdot Z_{CORR\_T1}^2}{c_1 + 2^{-(r_{ADC}-1)} \cdot c_6 \cdot Z_{CORR\_T1} + 2^{-2(r_{ADC}-1)} \cdot c_7 \cdot Z_{CORR\_T1}^2}$$

$$Y \in [0; 1)$$

$$P = Y \cdot (1 - 2^{-15} \cdot c_2 - 2^{-15} \cdot c_3) + 2^{-15} \cdot c_2 \cdot Y^2 + 2^{-15} \cdot c_3 \cdot Y^3$$

$$P \in [0; 1)$$

|                |                                                                                                          |
|----------------|----------------------------------------------------------------------------------------------------------|
| $r_{ADC}$      | Resolution of A/D-Conversion                                                                             |
| $Z_{CORR\_p}$  | Raw A/D-result for pressure (auto-zero compensated)                                                      |
| $Z_{CORR\_T1}$ | Raw A/D-result for temperature 1 (auto-zero compensated)                                                 |
|                | Conditioning coefficients stored in EEPROM Register 0 to 7; $c_i \in [-2^{15}; 2^{15})$ complement on 2: |
| $c_0$ ...      | Bridge Offset                                                                                            |
| $c_1$ ...      | Gain                                                                                                     |
| $c_2$ ...      | Non-linearity 2 <sup>nd</sup> order                                                                      |
| $c_3$ ...      | Non-linearity 3 <sup>rd</sup> order                                                                      |
| $c_4$ ...      | Temperature coefficient Bridge Offset 1 <sup>st</sup> order                                              |
| $c_5$ ...      | Temperature coefficient Bridge Offset 2 <sup>nd</sup> order                                              |
| $c_6$ ...      | Temperature coefficient Gain 1 <sup>st</sup> order                                                       |
| $c_7$ ...      | Temperature coefficient Gain 2 <sup>nd</sup> order                                                       |

The first equation compensates the Offset and fits the Gain including its temperature dependence. The non-linearity is then corrected regarding the intermediate result Y. The result of these equations is a non-negative 15 Bit value P for the measurand (e.g. pressure) in the range [0; 1). This value P is continuously written to the output register of the digital serial interface during measurement cycle.

Note that the conditioning coefficients  $c_i$  are positive or negative values in complement on two.

### 2.3 Correction Formula for Temperature

There is also a possibility to condition the raw temperature value  $Z_{CORR\_T1/2}$  regarding Offset, Gain and Non-linearity. The result of this conditioning is a non-negative 15 Bit value  $T_{1/2}$  for temperature in the range [0; 1). This value  $T_{1/2}$  is continuously written to the output register of the digital serial interface during measurement cycle if output is enabled. Note that into the conditioning of measurand signal (chapter 2.2) the raw A/D-result  $Z_{CORR\_T1}$  for temperature is included.

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⇒ Range definition of inputs

$$Z_{CORR\_Ti} \in [-2^{r_{ADC}-1}; 2^{r_{ADC}-1}) \quad t_i \in [-2^{15}; 2^{15})$$

⇒ Conditioning Equations

$$Y_t = \frac{Z_{CORR\_Ti} + t_0}{t_1} \quad Y_t \in [0; 1)$$

$$T_i = Y_t \cdot (1 - 2^{-15} \cdot t_2) + 2^{-15} \cdot t_2 \cdot Y_t^2 \quad T_i \in [0; 1)$$

|                |                                                                                                              |
|----------------|--------------------------------------------------------------------------------------------------------------|
| $r_{ADC}$      | Resolution of A/D-Conversion                                                                                 |
| $Z_{CORR\_Ti}$ | Raw A/D-result temperature $T_1/T_2$ (auto-zero compensated)                                                 |
|                | Conditioning coefficients stored in EEPROM Register 10 to 15; $t_i \in [-2^{15}; 2^{15})$ complement on two: |
| $t_0 \dots$    | Temperature Offset                                                                                           |
| $t_1 \dots$    | Gain                                                                                                         |
| $t_2 \dots$    | Non-linearity 2 <sup>nd</sup> order                                                                          |

The first equation compensates the offset and fits the gain. The nonlinearity is then corrected regarding the intermediate result  $Y_t$ .

Note that the conditioning coefficients  $t_i$  are positive or negative values in complement on two.

## 2.4 Limiting the Output

The conditioning results for measurand P and temperature  $T_{1/2}$  can be output via different channels. For every channel an individual set of minimum and maximum limits is defined in EEPROM in consideration of the resolution of the chosen output channel.

⇒ Limitation

$$P_{out}(P > I_{max}) = I_{max}$$

$$P_{out}(P < I_{min}) = I_{min}$$

$$T_{out}(T > I_{max}) = I_{max}$$

$$T_{out}(T < I_{min}) = I_{min}$$

$$P_{out} \in [I_{min}; I_{max})$$

$$T_{out} \in [I_{min}; I_{max})$$

|                 |                                                                                                        |
|-----------------|--------------------------------------------------------------------------------------------------------|
|                 | Limits stored in EEPROM Register 8, 9 or 16, 17; $I_{min/max} \in [0; 2^{\text{Output-Resolution}})$ : |
| $I_{min} \dots$ | Lower output limit                                                                                     |
| $I_{max} \dots$ | Upper output limit                                                                                     |

Note that the limits  $I_{min}$ ,  $I_{max}$  must match the resolution of the regarded analog output. That means that the limits for the analog voltage output (resolution 11 Bit) have to be defined in the range  $[0; 2^{11})$ , the limits for the PWM output (maximum resolution 12 Bit) have to be defined in the range  $[0; 2^{12})$ .

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### 3. OUTPUTS

#### 3.1 General Description

The ZMD31050 offers a variety of output-modes via two output-channels for the conditioning results of measurand (e.g. pressure) and temperature signals. There are Analog Voltage Out, (4 to 20) mA Current Loop and two PWM-OUT. The three possible result-values for the measurand and the two temperatures can be switched arbitrary to these different outputs by EEPROM programming (CFGCYC:ACOS1, CFGCYC:ACOS2, see chapter 5.3). Furthermore, two ALARM-OUT are available which work only regarding the measurand value. Table 2 shows accessibility of result-values to the available outputs.

| Output                 | Accessible with |               |               | Pin  |
|------------------------|-----------------|---------------|---------------|------|
|                        | Measurand       | Temperature 1 | Temperature 2 |      |
| Analog voltage output  | yes             | yes           | yes           | OUT  |
| 4 to 20mA current loop | yes             | yes           | yes           | OUT  |
| PWM1                   | yes             | yes           | yes           | IO1  |
| PWM2                   | yes             | yes           | yes           | OUT  |
| Alarm 1 *              | yes             | -             | -             | IO1  |
| Alarm 2 *              | yes             | -             | -             | IO2* |

**Table 2:** Accessibility of outputs

The possible combinations of the available outputs are limited by the used pins OUT, IO1 and IO2. Consequently an output mode using the pin OUT can only be combined with an output mode using the pin IO1 or IO2 or both. The configuration is done by EEPROM programming (CFGOUT:COS1, CFGOUT:PMIO1, CFGOUT:PMIO2, see chapter 0). Table 3 shows all possible combinations of available outputs.

The attached result-value (measurand or temperature) is updated always when a new value is calculated using the conditioning equations (see chapter 2). In addition the calculated values are also available as digital values at the digital serial interface.

\* Application access to Port IO2 during active ZACwire-Mode (ZACwire Start Window or active "OWI for ever"-Mode) is not possible. ALARM function is not applicable, if an AD-Resolution lower than 13bit and second order is adjusted.

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| CFGOUT - EEPROM address<br>1B/27 (see chapter 5.3) |       |       | Pin                       |         |         |
|----------------------------------------------------|-------|-------|---------------------------|---------|---------|
| COS1                                               | PMIO1 | PMIO2 | OUT                       | IO1     | IO2     |
| 00                                                 | 0d    | 0     | Analog voltage output     | -       | -       |
| 01                                                 | 0d    | 0     | (4 to 20) mA current loop | -       | -       |
| 10                                                 | 0d    | 0     | PWM2                      | -       | -       |
| 00                                                 | 10    | 0     | Analog voltage output     | PWM1    | -       |
| 01                                                 | 10    | 0     | (4 to 20) mA current loop | PWM1    | -       |
| 10                                                 | 10    | 0     | PWM2                      | PWM1    | -       |
| 00                                                 | 11    | 0     | Analog voltage output     | Alarm 1 | -       |
| 01                                                 | 11    | 0     | (4 to 20) mA current loop | Alarm 1 | -       |
| 10                                                 | 11    | 0     | PWM2                      | Alarm 1 | -       |
| 00                                                 | 0d    | 1     | Analog voltage output     | -       | Alarm 2 |
| 01                                                 | 0d    | 1     | (4 to 20) mA current loop | -       | Alarm 2 |
| 10                                                 | 0d    | 1     | PWM2                      | -       | Alarm 2 |
| 00                                                 | 10    | 1     | Analog voltage output     | PWM1    | Alarm 2 |
| 01                                                 | 10    | 1     | (4 to 20) mA current loop | PWM1    | Alarm 2 |
| 10                                                 | 10    | 1     | PWM2                      | PWM1    | Alarm 2 |
| 00                                                 | 11    | 1     | Analog voltage output     | Alarm 1 | Alarm 2 |
| 01                                                 | 11    | 1     | (4 to 20) mA current loop | Alarm 1 | Alarm 2 |
| 10                                                 | 11    | 1     | PWM2                      | Alarm 1 | Alarm 2 |

**Table 3:** Available combinations of outputs

(d ... don't care)

### 3.2 PWM Output

The ZMD31050 offers two PWM outputs at pin OUT and at pin IO1.

The PWM works synchronous

- To the internal oscillator frequency ( $f_{CLK} = 1 / 2 / 4 \text{ MHz}$ , CFGAPP:OSCF) or
- To an external frequency ( $f_{EXT} = (2 \dots 4) \text{ MHz}$ , CFGAPP:ECLKE = 1).

Using an external frequency ( $f_{EXT}$ ) it has to be supplied via pin IN3. Furthermore, it is possible to divide the used frequency by 2, 4 or 8 (CFGOUT:PWMD).

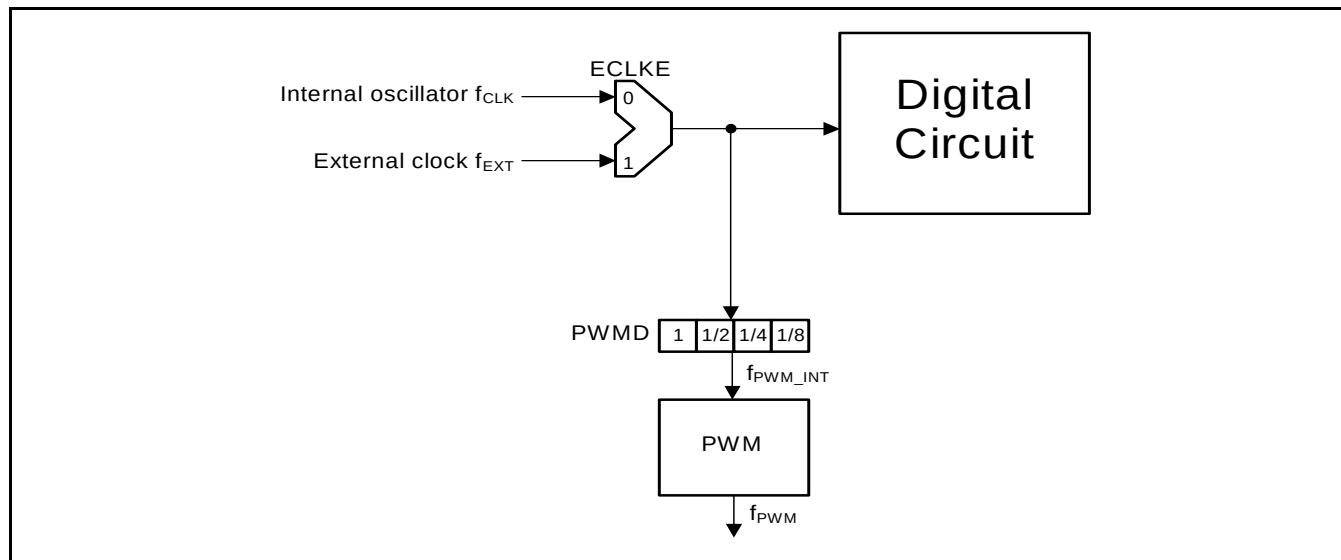
Fig. 2 illustrates the influence of the EEPROM configuration to the internal used frequencies.

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**Fig. 2:** Possible frequency supply for PWM output

The resolution  $r_{PWM}$  for PWM output (9 / 10 / 11 / 12 Bit) has to be defined again by EEPROM programming (CFGOUT:PWMRES). The resulting frequency of PWM output depends on this resolution and on the internal used frequency.

$$f_{PWM} = f_{PWM\_INT} / 2^{r_{PWM}}$$

This results in frequencies for the PWM output as listed in Table 4. Note that the chosen frequency  $f_{CLK}$  respectively  $f_{EXT}$  also influences the A/D-conversion rate.

| $f_{CLK}$ or $f_{EXT}$ | $d_{PWM}$<br>(CFGOUT:PWMD) | $r_{PWM}$ (CFGOUT:PWMRES) |          |          |        |
|------------------------|----------------------------|---------------------------|----------|----------|--------|
|                        |                            | 9 Bit                     | 10 Bit   | 11 Bit   | 12 Bit |
| 4 MHz                  | 1                          | 7.8 kHz                   | 3.9 kHz  | 1.95 kHz | 975 Hz |
| 4 MHz                  | 1/2                        | 3.9 kHz                   | 1.95 kHz | 975 Hz   | 490 Hz |
| 4 MHz                  | 1/4                        | 1.95 kHz                  | 975 Hz   | 490 Hz   | 245 Hz |
| 4 MHz                  | 1/8                        | 975 Hz                    | 490 Hz   | 245 Hz   | 122 Hz |
| 2 MHz                  | 1                          | 3.9 kHz                   | 1.95 kHz | 975 Hz   | 490 Hz |
| 2 MHz                  | 1/2                        | 1.95 kHz                  | 975 Hz   | 490 Hz   | 245 Hz |
| 2 MHz                  | 1/4                        | 975 Hz                    | 490 Hz   | 245 Hz   | 122 Hz |
| 2 MHz                  | 1/8                        | 490 Hz                    | 245 Hz   | 122 Hz   | 61 Hz  |
| 1 MHz                  | 1                          | 1.95 kHz                  | 975 Hz   | 490 Hz   | 245 Hz |
| 1 MHz                  | 1/2                        | 975 Hz                    | 490 Hz   | 245 Hz   | 122 Hz |
| 1 MHz                  | 1/4                        | 490 Hz                    | 245 Hz   | 122 Hz   | 61 Hz  |
| 1 MHz                  | 1/8                        | 245 Hz                    | 122 Hz   | 61 Hz    | 30 Hz  |

**Table 4:** Available frequencies of PWM output

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The result-value for PWM output is calculated by using the limiting equations (chapter 2.4) and is in the range  $[I_{\min}, I_{\max}]$ . These limits  $I_{\min}$ ,  $I_{\max}$  are defined in the EEPROM registers 8, 9 for PWM2 (Pin OUT) and 16, 17 for PWM1 (Pin IO1), respectively. It has to match the chosen resolution  $r_{\text{PWM}}$  for PWM output, i.e. the limits have to be defined in the range  $[0, 2^{r_{\text{PWM}}}]$ .

The resulting PWM signal for an output value OUT of measurand (e.g. pressure p), temperature T1 or temperature T2 has than a duty cycle  $dc_{\text{PWM}}$  of

$$dc_{\text{PWM}} = (\text{OUT} + 1) / 2^{r_{\text{PWM}}}$$

Furthermore, the polarity of the PWM output signal can be defined by EEPROM programming (CFGOUT:PWMPOL), i.e. this defines if a PWM clock cycle starts with high or low level.

### 3.3 Alarm Tasks

ZMD31050 offers two independent configurable alarm tasks (timer & comparator), which can be used as programmable switch output. The pins IO1 and IO2 are used as output for these ALARM tasks. The <OUT> output signal, defined by CFGCYC:ACOS1, is used as input/source for both alarm tasks.

The behaviour of the alarm output can be configured by 4 parameters for both alarms:

- Switch on threshold  $A_{\text{ON}}$
- Hysteresis  $A_{\text{HYST}}$  - is used for calculation of switch off threshold  $A_{\text{OFF}} = A_{\text{ON}} - A_{\text{HYST}}$
- Switch on delay  $A_{\text{td}_{\text{ON}}}$  & Switch off delay  $A_{\text{td}_{\text{OFF}}}$

Alarm task are realized 100%ly in firmware of 31050, so the NOM (running signal conditioning loop) is necessary for operation of alarm outputs. The programmed switch on/off threshold is compared with the current limited signal conditioning result for processing the alarm task.

#### 3.3.1 Alarm Outputs and Processing Task

The activation of programmable IOs is scheduled in 3 tasks:

- PowerOn and EEPROM-signatur calculation: approx. 2ms  
=> Alarm output IO1/IO2 is tristate.
- Processing of Start-procedure  
=> Alarm output IO1/IO2 is in programmed "inactive" state (CFGOUT:APO)
- Alarm operation depending of source and alarm configuration (registers: 0x16...0x21)  
=> IOs are working like programmed

Start procedure consist of 4...7 measurement tasks (PAZ, P, T1, T1AZ, T2, T2AZ, CMV - refer also datasheet "ZMD31050\_DS\_Rev\_\*.pdf" chapter 1.3.3). Every of these tasks needs determined conversion time defined by ADC-adjustment (ADC-resolution, ADC-order, ADC-clock divider, clock frequency – refer "ZMD31050\_bandwidth\_calculation\_Rev\*.xls") in range of 125µs to 100ms.

ALARM tasks aren't applicable, if an AD-Resolution lower than 13bit and second order is adjusted, otherwise a timing error is messaged.

#### 3.3.2 Alarm Configuration

The alarm output is set (to HIGH), if the programmed threshold  $A_{\text{ON}}$  is exceeded and reset, if the value is less than resulting threshold  $A_{\text{OFF}}$  (CFGOUT:APO[1]=1). CFGOUT:APO[0]=0 inverts this behaviour.

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|                                              |                                                                                                                                                                                                                                    |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Switch on threshold $A_{ON}$                 | $A_{ON} [\%] = \text{ALARM\_REG1} * 100 / 65536$ <p>with <b>ALARM_REG1</b> = RegNr "0x10"/"0x13" for Alarm1/2 (refer chapter 5.2)</p>                                                                                              |
| Switch off threshold $A_{OFF}$               | $A_{OFF} [\%] = A_{ON} [\%] - \text{ALARM\_REG2} * 65536 / 100$ <p>with <b>ALARM_REG2</b> = RegNr "0x11"/"0x14" for Alarm1/2 (refer chapter 5.2)</p>                                                                               |
| Switch on/off delay $A_{tdON}$ / $A_{tdOFF}$ | $A_{tdON} = \text{ALARM\_REG3}[7:0] * 4 * \text{AD\_ConversionTime}$ $A_{tdOFF} = \text{ALARM\_REG3} [15:8] * 64 * \text{AD\_ConversionTime}$ <p>with <b>ALARM_REG3</b> = RegNr "0x12"/"0x15" for Alarm1/2 (refer chapter 5.2)</p> |

**Table:** Alarm Parameter Calculation

### 3.3.3 Window Mode

If window alarm mode is active (CFGOUT:AWME=1) the thresholds for alarm1/2 define the limits of the window, hysteresis and ON/OFF delays work in the same way as described above. The both ALARM outputs are invert to one other. It is not important which limit is higher. When the input value is inside the window limits the ALARM1 shows:

- HIGH if programmed: ALARM1=High active & ALARM2=High active
- LOW if programmed: ALARM1=High active & ALARM2=Low active
- HIGH if programmed: ALARM1=Low active & ALARM2=Low active
- LOW if programmed: ALARM1=Low active & ALARM2=High active

All these values have to be programmed in the EEPROM.

The files "ALARM1.31050" and "ALARM2.31050" show how it works for a threshold 2.5V (Alarm1, low active) and 3.7V (Alarm2, low active) and the file "ALARM\_WINDOW.31050" shows how it works with those thresholds as window-limits.

### 3.4 Digital Readout

ZMD31050 supports digital readout also in NOM, three 16 bit SIF output registers can be read out. The count of registers to be read out is defined in configuration register CFGSIF and by setting of bits SIFOUTP, SIFOUTT1 & SIFOUT2. Read output result is a non limited 15bit value, the MSB is always "0". The MSB is the error identification bit, in case of an error these bit is set.

The output can be read out continuously (means without stop condition). Than the content of registers like programmed (CFGSIF) is sent in a loop. The content of SIF readout registers is refreshed, if a new conditioning result is valid. The SIF state machine guarantees, that a running readout operation is not disturbed by refreshing of the readout data.

Refer chapter 4 for a detailed description of SIF and used communication protocols.

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### 4. SERIAL DIGITAL INTERFACE

#### 4.1 General Description

The ZMD31050 includes a serial digital interface, which is able to communicate using three different communication protocols – I<sup>2</sup>C, SPI and a ZACwire™ (one wire - OWI) communication (see Fig. 3). The serial digital interface allows the programming of the EEPROM to configure the application mode of the ZMD31050 and to calibrate the conditioning equation. Furthermore it makes possible the read out of the conditioning result of measurand and both temperatures as digital 15 Bit values. The ZMD31050 works always as slave. The used communication protocol has to be chosen by programming EEPROM. There are also commands to change the valid communication mode. Only one communication protocol is valid at one time. The implemented commands are available in all communication modes but are divided in two sets with different validity (refer chapter 1.3.3 and chapter 4.4). Commands, which change the configuration of the device, are suppressed in “Normal Operation Mode” (NOM) and are available only after changing to “Command Mode” (CM). A non-configured device, identified by a CRC Error regarding EEPROM contents, starts up in a special mode so that communication by any of the available protocols is possible. After power-on a 20ms-time-slot is opened to start OWI-communication (OWI-start-window). If no OWI-communication is detected, the serial digital interface changed in a mode in which it can receive commands in I<sup>2</sup>C-mode as well as in SPI-mode. If it is necessary to read out data from a non-configured device (transmission by slave), the used communication protocol has to be explicitly defined by sending a certain command. Note that the ZMD31050 also do not send acknowledges in non-configured mode if I<sup>2</sup>C-communication is used. If the ZMD31050 receives a valid command at the serial digital interface the measurement cycle is interrupted because the internal micro-controller has to execute the requested command routine. An exception is a read-request. The ZMD31050 answers without interrupting the measurement cycle. This makes possible the read-out of the digital conditioning results during measurement cycle. A command consists of an address byte and a command byte. Additionally the commands for writing the EEPROM or its mirror in the RAM enclose two data bytes. This is independent from used communication protocol. To read data from the ZMD31050 (e.g. EEPROM contents) usually a certain command has to be sent to transfer this data into the output register of the serial interface. Thereafter the READ command consisting of the address byte with the read bit set is used to get this data. The data are transmitted continuously repeated as long as the master send the clock and do not abort by generating a stop condition. Again this is independent from used communication protocol. During the measurement cycle the ZMD31050 transfers the conditioning results into the output registers of the serial digital interface. There are three registers for pressure and both temperatures. The activation of these registers and consequently of the transmitted data have to be set by EEPROM programming. The results are sent in the sequence pressure, temperature 1 and temperature 2 according to the register activation if the master sends a read-request.

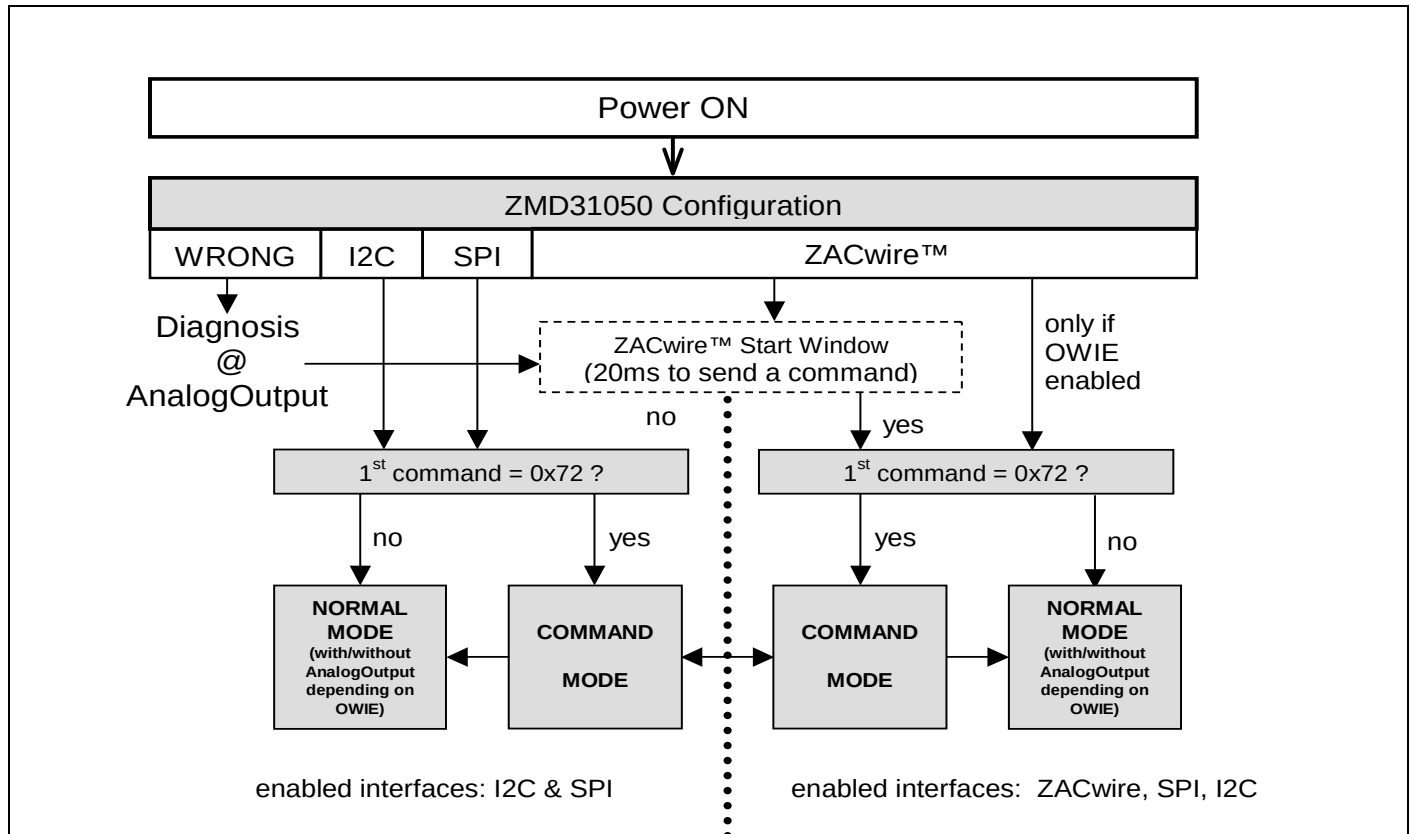
Communication is controlled by internal clock frequency. Internal clock frequency  $f_{CLK}$  has to be in minimum 5 times higher than communication clock frequency.

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**Fig. 3:** Possible Communication Configuration

### 4.2 I<sup>2</sup>C Protocol

For I<sup>2</sup>C communication a data line (SDA) and a clock line (SCL) are required. The I<sup>2</sup>C protocol used is defined as follows:

- **Idle period**

During inactivity of the bus SDA and SCL are pulled-up to supply voltage VDDA.

- **Start condition**

A high to low transition on SDA while SCL is at high level indicates a start condition. Every command has to be initiated by a start condition sent by a master. A master can always generate a start condition.

- **Stop condition**

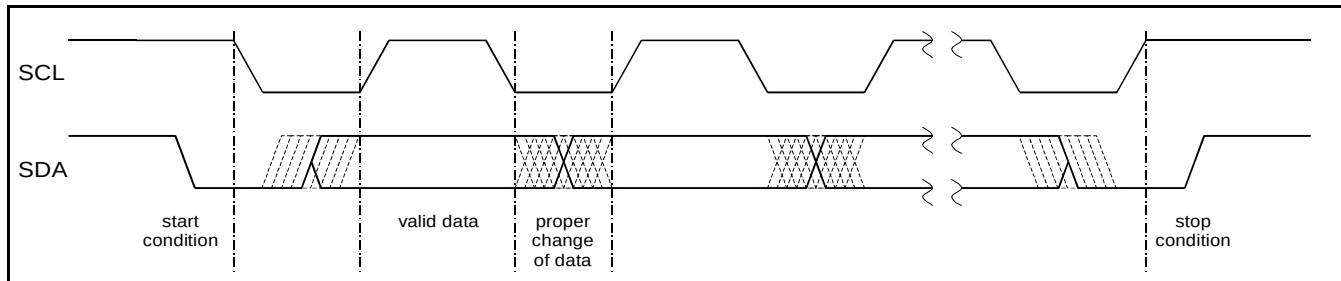
A low to high transition on SDA while SCL is at high level indicates a stop condition. A command has to be closed by a stop condition to start processing the command routine inside the IC.

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**Fig. 4:** Principles of I<sup>2</sup>C protocol

- **Valid data**

Data is transmitted in Bytes (8 Bits) starting with the most significant bit (MSB). Each byte transmitted is followed by an acknowledge bit. Transmitted bits are valid if after a start condition SDA keeps at constant level during high period of SCL. The SDA level has to change only when clock signal at SCL is low.

- **Acknowledge**

Acknowledge after transmitted byte is obligatory. The master must generate an acknowledge related clock pulse. The receiver (slave or master) pulls-down the SDA line during acknowledge clock pulse. If no acknowledge is generated by the receiver a transmitting slave will set inactive. A transmitting master can abort the transmission by generating a stop condition and may repeat the command.

A receiving master must signal the end of transfer to the transmitting slave by not generating an acknowledge related clock pulse at SCL.

The ZMD31050 as a slave changes to inactive interface mode during processing internal command routines started by a previously sent command.

- **Addressing**

Every slave connected to the I<sup>2</sup>C-bus responds to a certain address. After generating the start condition the master sends the address byte containing a 7bit-address followed by a data direction bit (R/W). A '0' indicates a transmission from master to slave (WRITE), a '1' indicates a data request (READ).

The addressed slave answers with an acknowledge, all other slaves connected with the I<sup>2</sup>C-bus ignore this communication.

The general ZMD31050 slave address is 0x78 (7bit). By EEPROM programming it is possible to allocate and activate an additional arbitrary slave address to every single device. In this case the device recognizes communication on both addresses, on the general one and on the activated one.

- **Write operation**

During transmission from master to slave (WRITE) after the address byte follows a command byte and, depending on the transmitted command, optional 2 data bytes. The internal micro-controller evaluates the received command and processes the related routine. The available commands are described in chapter 4.5.

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|         |                 |               |       |       |              |          |                           |       |                           |       |                |
|---------|-----------------|---------------|-------|-------|--------------|----------|---------------------------|-------|---------------------------|-------|----------------|
|         |                 |               |       |       |              | optional |                           |       |                           |       |                |
|         | start condition | slave address | R/W 0 | A     | command byte | A        | 1 <sup>st</sup> data byte | A     | 2 <sup>nd</sup> data byte | A     | stop condition |
| send by | master          |               |       | slave | master       | slave    | master                    | slave | master                    | slave | master         |

**Fig. 5:** Write operation  $I^2C$

#### • Read operation

After a data request from master to slave by sending an address-byte including a set data direction bit the slave answers by sending data from the activated interface output registers. The master must generate the transmission clock on SCL, the acknowledges after each data byte (except after the last one) and finally the stop condition.

A data request is answered by the interface module itself and does consequently not interrupt the current process of the internal micro-controller. The data in the activated registers is sent continuously until a stop condition is detected, after transmitting all available data the slave starts repeating the data.

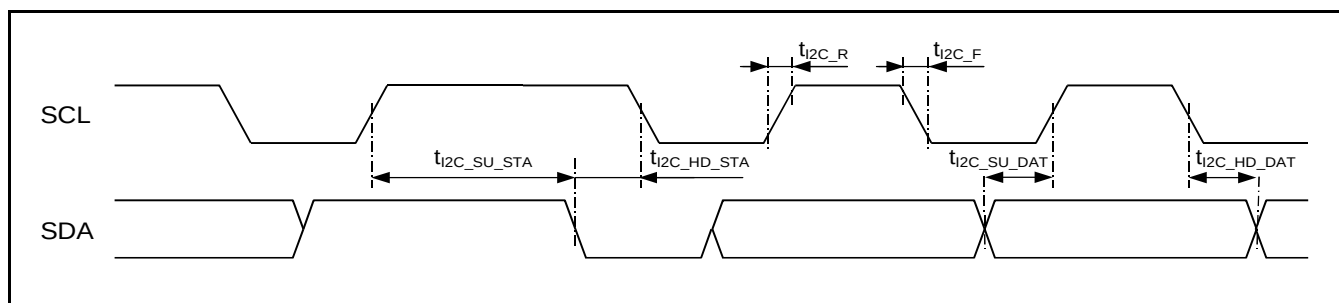
During running measurement cycle data is incessantly updated with conditioning results. To get other data from slave (e.g. EEPROM contents) usually a certain command has to be sent before the data request to initiate the transfer of this data to the interface output registers. This command does interrupt the current process of the internal microprocessor and consequently also a running measurement cycle.

|         | start condition | slave address | R/W 1 | A     | 1st data byte | A      | 2nd data byte | A      | ... nth data byte | stop condition |
|---------|-----------------|---------------|-------|-------|---------------|--------|---------------|--------|-------------------|----------------|
| sent by | master          |               |       | slave | slave         | master | slave         | master | slave             | Master         |

loop (only pressure readout in this case)

optional

**Fig. 6:** Read operation - data request  $I^2C$  (nth data byte is the last required data byte, readout can be processed in loop without resend initialisation by master)

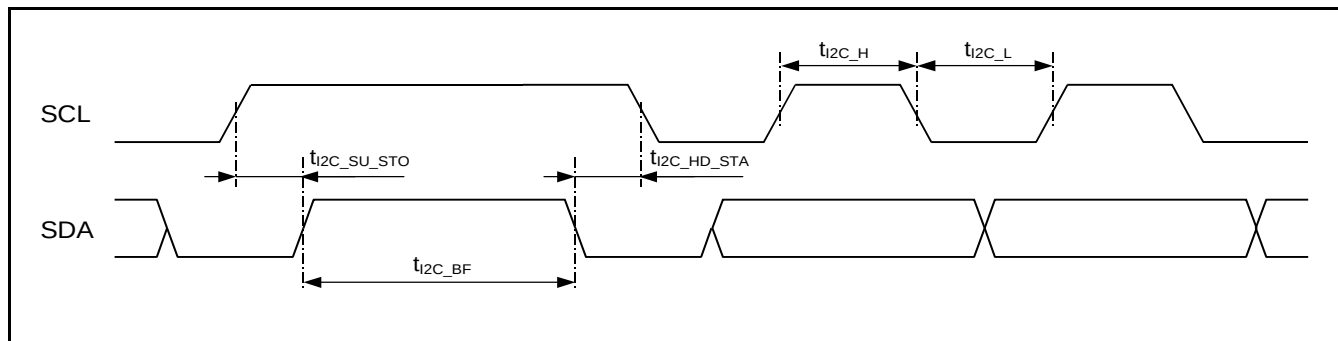


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**Fig. 7:** Timing I<sup>2</sup>C protocol

| Nr. | Parameter                                    | Symbol                  | min | typ | max | Unit | Conditions              |
|-----|----------------------------------------------|-------------------------|-----|-----|-----|------|-------------------------|
| 1   | SCL Clock frequency <sup>1</sup>             | f <sub>SCL</sub>        |     |     | 400 | kHz  | f <sub>CLK</sub> ≥ 2MHz |
| 2   | Bus free time betw. start and stop condition | t <sub>I2C_BF</sub>     | 1.3 |     |     | μs   |                         |
| 3   | Hold time start condition                    | t <sub>I2C_HD_STA</sub> | 0.6 |     |     | μs   |                         |
| 4   | Setup time repeated start condition          | t <sub>I2C_SU_STA</sub> | 0.6 |     |     | μs   |                         |
| 5   | Low period SCL/SDA                           | t <sub>I2C_L</sub>      | 1.3 |     |     | μs   |                         |
| 6   | High period SCL/SDA                          | t <sub>I2C_H</sub>      | 0.6 |     |     | μs   |                         |
| 7   | Data hold time                               | t <sub>I2C_HD_DAT</sub> | 0   |     |     | μs   |                         |
| 8   | Data setup time                              | t <sub>I2C_SU_DAT</sub> | 0.1 |     |     | μs   |                         |
| 9   | Rise time SCL/SDA                            | t <sub>I2C_R</sub>      |     |     | 0.3 | μs   |                         |
| 10  | Fall time SCL/SDA                            | t <sub>I2C_F</sub>      |     |     | 0.3 | μs   |                         |
| 11  | Setup time stop condition                    | t <sub>I2C_SU_STO</sub> | 0.6 |     |     | μs   |                         |
| 12  | Noise interception SDA                       | t <sub>I2C_NI</sub>     |     |     | 50  | ns   | Spikes are suppressed   |

**Table 5** Timing I<sup>2</sup>C protocol

### 4.3 OWI – OneWire Digital Interface

The ZMD31050 employs ZMD's OWI single wire digital interface. It combines a simple and easy protocol adaptation with a cost saving pin sharing. Both the analog voltage output and this digital interface (calibration and/or digital output value) occur over the same pin. An advantage of ZMD's OWI output signal capability is, that it enables "end of line" calibration – no additional pins are required to digitally calibrate a finished assembly. This reduces cost and complexity. The OWI interface is intended for calibration use only, although the calibrated output signal is also available via this interface. The master slave configuration is shown below.

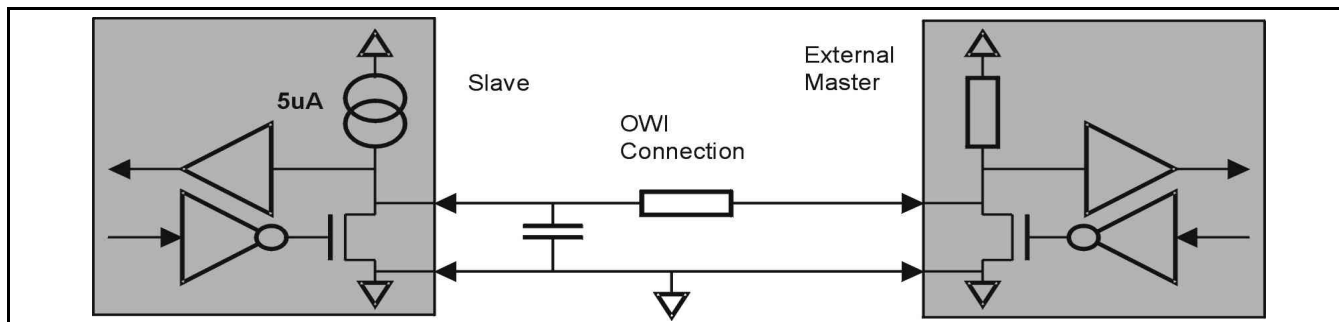
<sup>1</sup> Internal clock frequency f<sub>CLK</sub> has to be in minimum 5 times higher than communication clock frequency.

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**Fig. 8:** Block scheme of OWI connection

Both devices are peers, however only the external device starts communication and requests data, in this sense it is referred to as master and the ZMD31050 as slave.

### 4.3.1 Properties and Parameters

Though OWI is designed as bilateral protocol, it is necessary for reasons of compatibility to use an address in the communication. After the start condition the master has to send an address-byte, consisting of a 7 bit slave address and a read/write-bit (0 – write, 1 – read). The slave-address of the ZMD31050 is 78<sub>hex</sub>, additionally 127 free programmable addresses (00<sub>hex</sub> to 7F<sub>hex</sub>) are available.

| Nr. | Parameter                 | Symbol                |     | Unit | Conditions                                                            |
|-----|---------------------------|-----------------------|-----|------|-----------------------------------------------------------------------|
| 1   | Pull up resistance master | R <sub>OWI_PU</sub>   | 330 | Ω    | refer datasheet chapter 5.5.2 for detailed description of calculation |
| 2   | OWI line resistance       | R <sub>OWI_LINE</sub> | 20  | Ω    |                                                                       |
| 3   | OWI load capacitance      | C <sub>OWI_LOAD</sub> | 4.7 | nF   |                                                                       |

**Table 6:** OWI interface parameters – worst case parameterization

### 4.3.2 Initializing OWI communication

There are two possibilities to start one wire communication. First the ZMD31050 opens a 20ms-time-slot after power-on – the one wire start window. If in this time-slot one wire communication is detected the device stays in the one wire mode until this is closed by sending a certain command. If no one wire communication occurs in the start window the interface leaves the one wire mode and changes to the configured interface mode I2C or SPI as programmed in EEPROM. The one wire start window can be suppressed by configuration (CFGSIIF:OWIWIND = 1) if no access via one wire communication is desired. Second the ZMD31050 can be configured by EEPROM programming so that OWI is enabled also in NOM during measurement cycle (CFGSIIF:OWIWIND = 0 and CFGSIIF:OWIE = 1). Then it is possible to read out the actual conditioning results for measurand and temperature via one wire communication (Pin OUT), the analog voltage output is disabled.

### 4.3.3 OWI Protocol

The OWI protocol used is defined as follows:

- **Idle period**

During inactivity of the bus OWI line is pulled-up to supply voltage VD<sub>DA</sub>.

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- **Start condition**

When OWI line is in idle mode a low pulse (return to high) with minimum width of 10 $\mu$ s indicates a start condition. Every command has to be initiated by a start condition sent by a master. A master can generate a start condition only when OWI line is in idle mode.

- **Stop condition**

The master finishes a transmission by changing back to high level (idle mode). Every command (refer "write operation") has to be closed by a stop condition to start processing the command. The master can interrupt a sending slave after data request (refer "read operation") by clamping the OWI line to low level for generating a stop condition.

No transition from low to high or from high to low (constant level) at OWI line for at least twice the period of last transmitted valid bit indicates a stop condition. A stop condition without considering last bit-time (secure stop condition) is generated at constant level at OWI line for more than 510 clocks of internal clock oscillator.

- **Valid data**

Data is transmitted in Bytes (8 Bits) starting with the most significant bit (MSB). Transmitted bits are recognized after a start condition at every transition from low to high at OWI line. The value of the transmitted bit depends on the duty ratio between high phase and high/low period (bit period,  $t_{OWI\_BIT}$ ). A duty ratio greater than 1/8 and less than 3/8 is detected as '0', a duty ratio greater than 5/8 and less than 7/8 is detected as '1'.

The bit period of consecutive bits may not change more than with factor of 2 because stop condition is detected in this case.

- **Addressing**

Though OWI is designed as bilateral protocol, the slave responds only to a certain address (compatibility to I<sup>2</sup>C protocol). After generating the start condition the master sends the address byte containing a 7bit-address followed by a data direction bit (R/W). A '0' indicates a transmission from master to slave (WRITE), a '1' indicates a data request (READ).

The general ZMD31050 slave address is 0x78<sub>hex</sub> (7bit). By EEPROM programming it is possible to allocate and activate an additional arbitrary slave address to every single device. In this case the device recognizes communication on both addresses, on the general one and on the activated one.

- **Write operation**

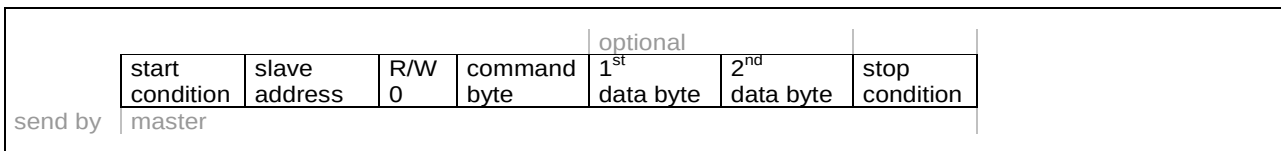
During transmission from master to slave (WRITE) after the address byte follows a command byte and, depending on the transmitted command, optional 2 data bytes. The internal microprocessor evaluates the received command and processes the related routine. The available commands are described in chapter 4.5.

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**Fig. 9:** Write operation ZACwire™

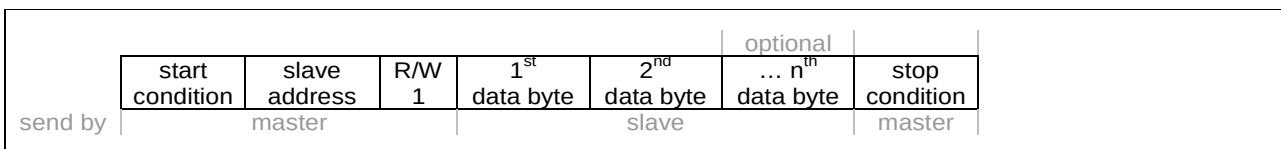
#### • Read operation

After a data request from master to slave by sending an address byte including a set data direction bit the slave answers by sending data from the activated interface output registers. The slave generates the data bits with a bit period equal to last received bit (R/W bit). The master has to generate a stop condition after receiving the requested data.

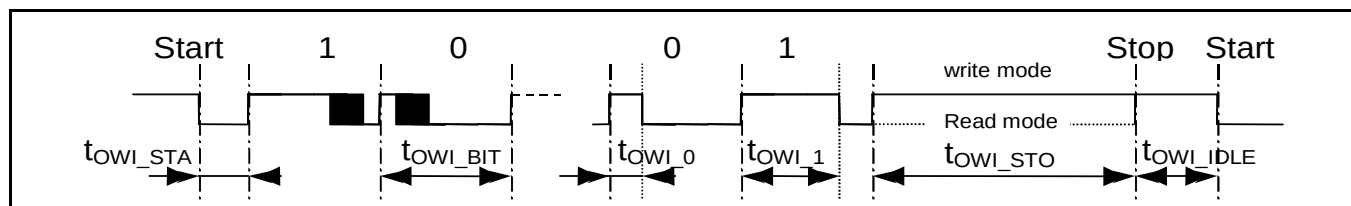
A data request is answered by the interface module itself and does consequently not interrupt the current process of the internal microprocessor.

The data in the activated registers is sent continuously until a stop condition is detected, after transmitting all available data the slave starts repeating the data.

During running measurement cycle data is incessantly updated with conditioning results. To get other data from slave (e.g. EEPROM contents) usually a certain command has to be sent before the data request to initiate the transfer of this data to the interface output registers. This command does interrupt the current process of the internal micro-controller and consequently also a running measurement cycle.



**Fig. 10:** Read operation - data request ZACwire™



**Fig. 11:** Timing ZACwire™ protocol

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| Nr. | Parameter                                      | Symbol              | min   | typ  | max   | Unit                               | Conditions                                        |
|-----|------------------------------------------------|---------------------|-------|------|-------|------------------------------------|---------------------------------------------------|
| 1   | Bus free time between start and stop condition | $t_{OWI\_IDLE}$     | 10    |      |       | $\mu\text{s}$                      |                                                   |
| 2   | Hold time start condition                      | $t_{OWI\_STA}$      | 10    |      |       | $\mu\text{s}$                      |                                                   |
| 3   | Bit period                                     | $t_{OWI\_BIT}$      | 20    |      | 100   | $\mu\text{s}$                      |                                                   |
| 4   | Duty ratio bit '0'                             | $t_{OWI\_0}$        | 0.125 | 0.25 | 0.375 | $t_{OWI\_BIT}$                     |                                                   |
| 5   | Duty ratio bit '1'                             | $t_{OWI\_1}$        | 0.625 | 0.75 | 0.875 | $t_{OWI\_BIT}$                     |                                                   |
| 6   | Hold time stop condition                       | $t_{OWI\_STO}$      | 2.0   | 250  |       | $t_{OWI\_BIT\_L}$<br>$\mu\text{s}$ | $t_{OWI\_BIT\_L}$ is bit period of last valid bit |
| 7   | Bit period deviation                           | $t_{OWI\_BIT\_DEV}$ | 0.75  | 1.0  | 1.25  | $t_{OWI\_BIT}$                     |                                                   |

**Table 7:** Timing ZACwire™ protocol

### 4.3.4 Using the ZACwire™ -Interface

For ZACwire™ -Communication the analog output pin OUT will be used, consequently ZACwire™ allows "end of line"-calibration, which offers an inexpensive and powerful overall sensor module calibration at the very end of manufacturing process. However, even if the ZACwire™ was designed mainly for calibration it may also be used to digitally read out the calibrated sensor signal continuously. These two communication tasks are supported in different step-up of the interface.

#### 1. "One-wire-start-window" - for calibration only

In that mode the IC listens during the first 20ms after Power On for communication. To open up the communication the START\_CM command including the slave address has to be sent. After that the output pin keeps the digital interface mode until the command START\_CYCL\_RAM / EEP or after Power On. The window can be suppressed by setting the lock bit "OWIWind" in EEPROM register 23/17<sub>hex</sub>. But this should be done as the last step of calibration only (because reactivating of this window is then only possible via I2C).

#### 2. "OWI for ever" - for calibration and for digital read out

Activate the ZACwire™ mode via start window (see 1.) and enable "OWI for NOM" Bit (OWIE-> "OWI for ever") with the write command to EEPROM register 23/17<sub>hex</sub>. It is now a calibration and also digital read out in ZACwire™-using mode possible. The configuration for the read out is determined in the configuration word CFGSIF (EEPROM register 23/17<sub>hex</sub>) with the SIFOUTP / SIFOUTT1 / SIFOUTT2 bits. The request for desired output values (P, T1, T2) is a read command to the slave address (e.g. to the slave address 78<sub>hex</sub> send F1<sub>hex</sub>). Depending on the configuration the IC sends for the first request the pressure value, if configured for the next the T1 value followed by the T2 value and starts again with the pressure. The master has to generate a stop condition after receiving the requested data.

### 4.3.5 Hints for using ZACwire™ with ZMD31050-Software and Application KIT 1

By using the RB<sup>IC</sup> Application Kit (KIT1) and the software the jumper JP1 has to be connect to "OWI to V<sub>OUT</sub>". To meet the 20ms start window select interface: OWI – ParallelPort-Kit. After pressing <ok> in the message window the START\_CM command will be sent continuously for 2++ seconds - you have approx. 2++ seconds to turn on the kit power supply. The kit communicates in ZACwire™ mode now.

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For writing data to the RAM use the “WRITE”-button in the “RAM-Register”- subwindow, if you don't want to start the Normal Operation Mode (NOM=continuous measurement). The “WRITE + CYC”-button in the main window (section “ASIC-Configuration”) starts the Normal Operation Mode (NOM), which concludes the ZACwire™ -access, if you have not activated the “OWI for ever”-mode.

### 4.4 Synchronous serial peripheral interface (SPI) Introduction

Using of SPI interface of ZMD31050 needs a special configuration in EEPROM to activate these communication protocol, which is also different from default configuration. It is necessary to have I2C or OWI communication access to ZMD31050 to write the SPI configuration into EEPROM. It is no I2C or OWI communication possible, if SPI configuration is written into EEPROM and activated.

During an SPI transfer, data is simultaneously transmitted (shifted out serially) and received (shifted in serially). A serial clock line synchronizes shifting and sampling of the information on the two serial data lines. A slave select line allows individual selection of a slave SPI device; slave devices those are not selected do not interfere with SPI bus activities. ZMD31050's SPI slave interface supports all combinations of clock phase (CPHA) and polarity (CPOL). Slave CPOL and CPHA has to be programmed to master adjustments before beginning of transmission. RAM/EEPROM register 0x17<sub>hex</sub> contains initialization of SPI interface:

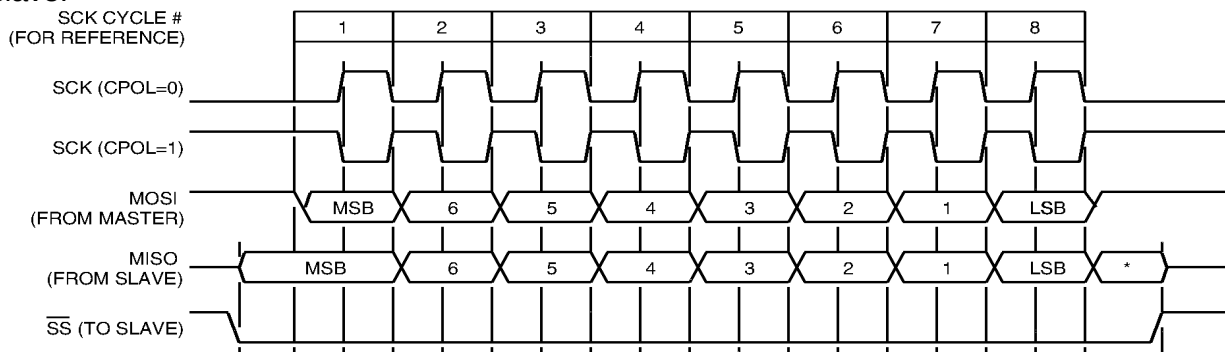
|                |                             |
|----------------|-----------------------------|
| SFGSIF: SIFMD  | = interface mode SPI or I2C |
| SFGSIF: SPICKP | = clock polarity CPOL       |
| SFGSIF: SPICKE | = clock phase CPHA          |

#### 4.4.1 SPI Clock Phase and Polarity Controls

Software can select any of four combinations of serial clock (SCK) phase and polarity using two bits in the SPI control register (SPCR). The clock polarity is specified by the CPOL control bit, which selects an active high or active low clock and has no significant effect on the transfer format. The clock phase (CPHA) control bit selects one of two fundamentally different transfer formats.

#### CPHA Equals Zero Transfer Format

Fig. 12 shows a timing diagram of an SPI transfer where CPHA is zero. Two waveforms are shown for SCK: one for CPOL equals zero and another for CPOL equals one. The diagram may be interpreted as a master or slave timing diagram since the SCK, master in/slave out (MISO), and master out/slave in (MOSI) pins are directly connected between the master and the slave. The MISO signal is the output from the slave, and the MOSI signals the output from the master. The /SS line is the slave select input to the slave.



**Fig. 12:** CPHA Equals Zero SPI Transfer Format – principle transfer illustration

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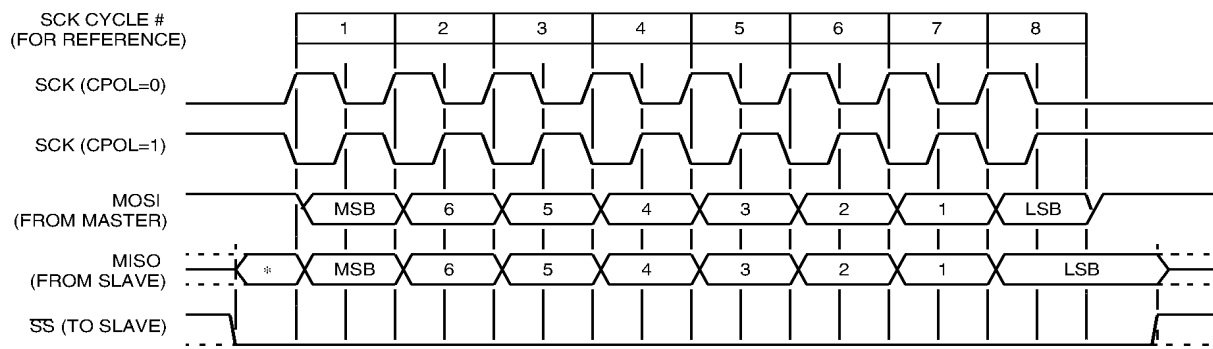
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### CPHA Equals One Transfer Format

Fig. 13 shows a timing diagram of an SPI transfer where CPHA is one. Two waveforms are shown for SCK: one for CPOL equals zero and another for CPOL equals one. The diagram may be interpreted as a master or slave timing diagram since the SCK, MISO, and MOSI pins are directly connected between the master and the slave. The MISO signal is the output from the slave, and the MOSI signal is the output from the master. The /SS line is the slave select input to the slave.



**Fig. 13:** CPHA Equals One SPI Transfer Format – principle transfer illustration

When CPHA equals zero, the /SS line must be negated and reasserted between each successive serial byte.

When CPHA equals one, the /SS line may remain active low between successive transfers (can be tied low at all times). This format is sometimes preferred in systems having a single fixed master and a single slave driving the MISO data line.

### 4.4.2 SPI Pin Signals

There are four I/O pin signals associated with SPI transfers: the SCK, the MISO data line, the MOSI data line, and the active low /SS pin. When the master initiates a transfer, eight clock cycles are automatically generated on the SCK pin. The SCK pin is an input and the clock signal from the master synchronizes the data transfer between the master and slave devices. Slave devices ignore the SCK signal unless the /SS pin is active low. In both the master and slave SPI devices, data is shifted on one edge of the SCK signal and is sampled on the opposite edge where data is stable. Edge polarity is determined by the SPI transfer protocol.

The MISO and MOSI data pins are used for transmitting and receiving serial data. At slave MOSI is the data input line, and MISO is the data output line. One by master selected slave device optionally drives data out its MISO pin to the MISO master pin. The automatic control of the direction of these pins makes reconfiguration through external logic unnecessary when a new device becomes the master.

The /SS pin behaves differently on master and slave devices. On a slave device, this pin is used to enable the SPI slave for a transfer. If the /SS pin of a slave is inactive (high), the device ignores SCK clocks and keeps the MISO output pin in the high-impedance state.

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#### 4.4.3 Beginning and Ending SPI Transfers

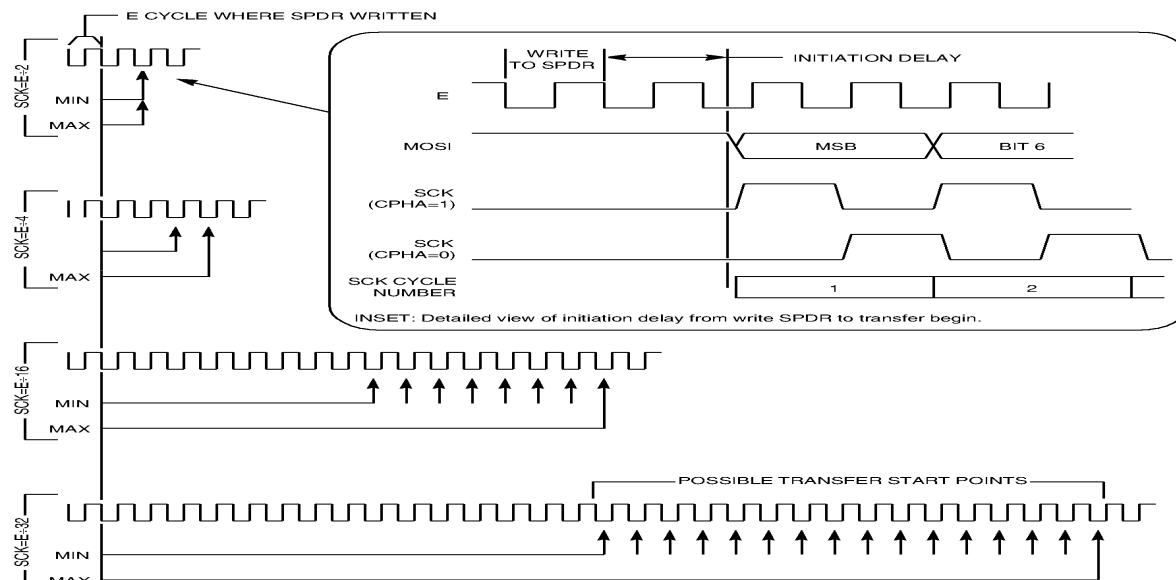
A transfer includes the eight SCK cycles plus an initiation period at the beginning and ending period of the transfer. The details of the beginning and ending periods depend on the CPHA format selected and whether the SPI is configured as a master or a slave. The initiation delay period is also affected by the SPI clock rate selection when the SPI is configured as a master.

It may be useful to refer to the transfer format illustrated in Fig. 12 and Fig. 13 to understand how the beginning and ending details fit into a complete transfer operation.

#### 4.4.4 Transfer Beginning Period (Initiation Delay)

All SPI transfers are started and controlled by a master SPI device. As a slave the transfer to begin with the first SCK edge or the falling edge of /SS, depending on the CPHA format selected. When CPHA equals zero, the falling edge of /SS indicates the beginning of a transfer. When CPHA equals one, the first edge on the SCK indicates the start of the transfer. In either CPHA format, a transfer can be aborted by taking the /SS line high, which causes the SPI slave logic and bit counters to be reset. The SCK rate selected has no effect on slave operations since the clock from the master is controlling transfers.

CPHA has no effect on the delay to the start of the transfer, but it does affect the initial state of the SCK signal. When CPHA equals zero, the SCK signal remains inactive for the first half of the first SCK cycle. When CPHA equals one, the first SCK cycle begins with an edge on the SCK line from its inactive to its active level. The SPI clock rate (selected by SPR[1:0]) affects the delay from the write to SPDR and the start of the SPI transfer (see Fig. 14). The internal SPI clock in the master is a free-running derivative of the internal MCU clock. Since the SPI clock is free-running, there is an uncertainty about when the write-operation to SPDR will occur relative to the slower SCK. This uncertainty causes the variation in the initiation delay shown in Fig. 14.



**Fig. 14:** Delay from Write SPDR to Transfer Start (Master)

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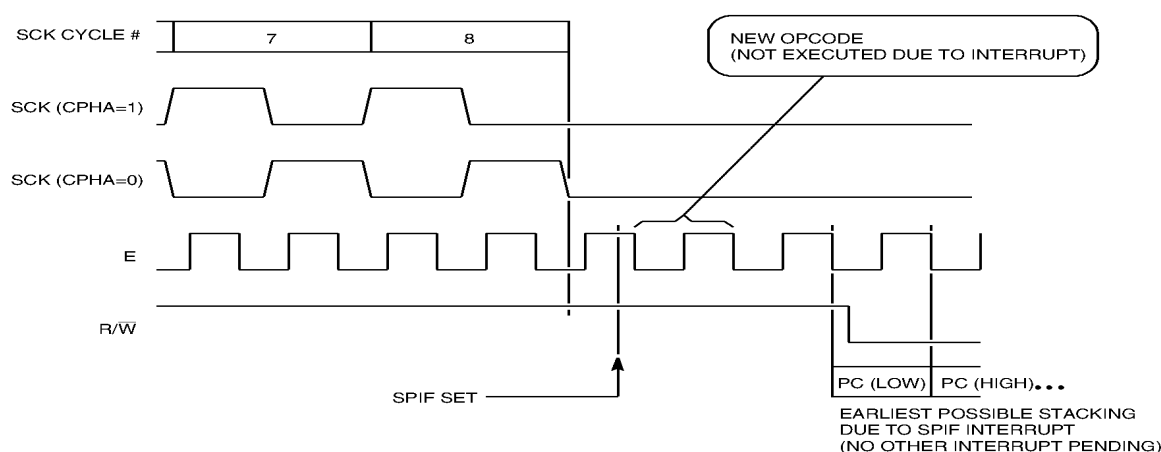
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#### 4.4.5 Transfer Ending Period

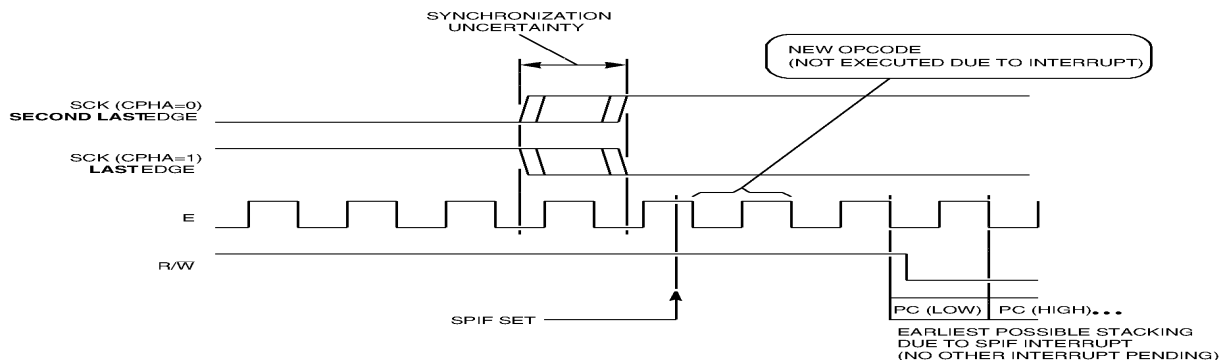
An SPI transfer is technically complete when the SPIF flag is set, but, depending on the configuration of the SPI system, there may be additional tasks. Because the SPI bit rate does not affect timing of the ending period, only the fastest rate will be considered in discussions of the ending period.

When the SPI is configured as a master, SPIF is set at the end of the eighth SCK cycle. When CPHA equals one, SCK is inactive for the last half of the eighth SCK cycle. Fig. 15 shows the transfer ending period for a master. The SCK waveforms in this figure show only the CPOL equals zero case, since clock polarity does not affect timing of the ending period.



**Fig. 15:** Transfer Ending for an SPI Master

When the SPI is operating as a slave, the ending period is different because the SCK line can be asynchronous to the MCU clocks of the slave and because the slave does not have access to as much information about SCK cycles as the master. For example, when CPHA equals one, where the last SCK edge occurs in the middle of the eighth SCK cycle, the slave has no way of knowing when the end of the last SCK cycle is. For these reasons, the slave considers the transfer complete after the last bit of serial data has been sampled, which corresponds to the middle of the eighth SCK cycle. A synchronization delay is required so the setting of the SPIF flag is properly positioned relative to the internal clock of the slave. Fig. 16 shows the ending period for a slave. The SCK waveforms in this figure show only the CPOL equals zero case, since clock polarity does not affect timing of the ending period.



**Fig. 16:** Transfer Ending for an SPI Slave

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When CPHA equals zero, there is a potential problem that can be avoided by proper software but is sometimes overlooked. The SPIF flag is set at the end of a transfer, but the slave is not permitted to write new data to the SPDR while the /SS line is still low. If the master device is busy, the /SS line to the slave can remain low longer than the slave expects.

### 4.5 Interface Commands

All implemented commands are available in all communication modes – I<sup>2</sup>C, SPI and ZACwire™. A received valid command interrupts the internal micro-controller and initiates the processing of a command routine.

| Command (HEX)                                                                                                       | Data   | Command        | Remarks                                                                                                                                                                                                                        | Processing time<br>System clock 2MHz |
|---------------------------------------------------------------------------------------------------------------------|--------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| 01                                                                                                                  |        | START_CYC_EEP  | Start measurement cycle including initialization from EEPROM                                                                                                                                                                   | 350µs                                |
| 02                                                                                                                  |        | START_CYC_RAM  | Start measurement cycle including initialization from RAM                                                                                                                                                                      | 220µs                                |
| 10 .. 1F                                                                                                            |        | READ_RAM0      | Read data from RAM address 00 .. 0F<br>Writes data from RAM to SIF Output Registers<br>Usually followed by Read operation                                                                                                      | 50µs                                 |
| 20 .. 2F                                                                                                            |        | READ_RAM1      | Read data from RAM address 10 .. 1F<br>Writes data from RAM to SIF Output Register<br>Usually followed by Read operation                                                                                                       | 50µs                                 |
| 30 .. 3F                                                                                                            |        | READ_EEP0      | Read data from EEPROM address 00 .. 0F<br>Writes data from EEPROM to SIF Output Register<br>Usually followed by Read operation                                                                                                 | 50µs                                 |
| 40 .. 4F                                                                                                            |        | READ_EEP1      | Read data from EEPROM address 10 .. 1F<br>Writes data from EEPROM to SIF Output Register<br>Usually followed by Read operation                                                                                                 | 50µs                                 |
| 5*-commands do not change EEPROM or RAM configuration, used for communication to devices with unknown configuration |        |                |                                                                                                                                                                                                                                |                                      |
| 50                                                                                                                  |        | CFG_SIF_TO_OWI | Configure SIF to Communication Mode OWI                                                                                                                                                                                        | 50µs                                 |
| 51                                                                                                                  |        | CFG_SIF_TO_SPI | Configure SIF to Communication Mode SPI                                                                                                                                                                                        | 50µs                                 |
| 55                                                                                                                  |        |                | 51 SET_SIF_2_SPI SPI (CKE = 0, CKP = 0)                                                                                                                                                                                        |                                      |
| 59                                                                                                                  |        |                | 55 SET_SIF_2_SPI SPI (CKE = 0, CKP = 1)                                                                                                                                                                                        |                                      |
| 5D                                                                                                                  |        |                | 59 SET_SIF_2_SPI SPI (CKE = 1, CKP = 0)                                                                                                                                                                                        |                                      |
|                                                                                                                     |        |                | 5D SET_SIF_2_SPI SPI (CKE = 1, CKP = 1)                                                                                                                                                                                        |                                      |
| 52                                                                                                                  |        | CFG_SIF_TO_I2C | Configure SIF to Communication Mode I <sup>2</sup> C                                                                                                                                                                           |                                      |
| 60                                                                                                                  | 2 Byte | SET_DAC        | Set Output DAC to value defined by data bytes [0x0 ... 0x7FF]<br>hint: only applicable for SPI and I2C communication mode.<br>In OWI Mode use the limit registers 0x8,0x9 followed by command "02" for the same functionality. | 100µs                                |
| 70                                                                                                                  |        | START_OM       | Start Open Mode<br>In Open Mode the command set is restricted!<br>Change to Command Mode is possible                                                                                                                           | 50µs                                 |
| 71                                                                                                                  |        | START_NOM      | Start Normal Operation Mode (NOM)<br>In Normal Operation Mode the command set is restricted!<br>Change to Command Mode or Open Mode is NOT possible                                                                            |                                      |
| 72                                                                                                                  |        | START_CM       | Start Command Mode (CM)<br>Complete command set!<br>Change to Open Mode or Normal Operation Mode is possible                                                                                                                   |                                      |

**Table 8:** Restricted command set for serial digital interface – commands are always valid

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During processing time the digital serial interface (SIF) is disabled, transmitted commands are ignored. The processing time depends from the internal system clock frequency, which is usually 2MHz but is changeable by EEPROM programming. The commands are divided in two sets with different validity. Commands, which change the configuration of the device, are ignored in NOM and are available only after changing to CM.

| Command (HEX)                                                                                                                         | Data   | Command         | Remarks                                                                                                        | Processing time<br>System clock 2MHz<br>/ EEPROM-Progr.<br>Steps |
|---------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 80 .. 8F                                                                                                                              | 2 Byte | WRITE_RAM0      | Write data to RAM address 00 .. 0F                                                                             | 50µs                                                             |
| 90 .. 9F                                                                                                                              | 2 Byte | WRITE_RAM1      | Write data to RAM address 10 .. 1F                                                                             | 50µs                                                             |
| A0 .. AF                                                                                                                              | 2 Byte | WRITE_EEP0      | Write data to EEPROM address 00 .. 0F                                                                          | 12.5ms / 1                                                       |
| B0 .. BF                                                                                                                              | 2 Byte | WRITE_EEP1      | Write data to EEPROM address 10 .. 1F                                                                          | 12.5ms / 1                                                       |
| C0                                                                                                                                    |        | COPY_EEP2RAM    | Copy Contents of EEPROM to RAM<br>Restores EEPROM Configuration in RAM                                         | 130µs                                                            |
| C3                                                                                                                                    |        | COPY_RAM2EEP    | Copy Contents of RAM to EEPROM<br>Stores RAM Configuration to EEPROM<br>(signature is copied without check!!!) | 400ms / 32                                                       |
| C8                                                                                                                                    |        | GET_EEP_SIGN    | Calculates EEPROM Signature and output to SIF Output Register                                                  | 150µs                                                            |
| C9                                                                                                                                    |        | GEN_EEP_SIGN    | GET_EEP_SIGN + write the Signature to EEPROM address 0x1D                                                      | 12.6ms / 1                                                       |
| CA                                                                                                                                    |        | GET_RAM_SIGN    | Calculates RAM Signature and output to SIF Output Register                                                     | 150µs                                                            |
| CB                                                                                                                                    |        | GEN_RAM_SIGN    | GET_RAM_SIGN + write the Signature to RAM address 0x1D                                                         | 150µs                                                            |
| CC                                                                                                                                    |        | CLEAR_EEP       | Clear EEPROM<br>Sets complete EEPROM to 0x0000                                                                 | 12.5ms / 1                                                       |
| CF                                                                                                                                    |        | ROM_VERSION     | Read ROM version and write it to SIF Output Register<br>Usually followed by Read operation                     | 50µs                                                             |
| All "D*" -Commands: used for calibration process, write raw conversion result to SIF Output Registers, do not effect to analog output |        |                 |                                                                                                                |                                                                  |
| D0                                                                                                                                    |        | START_AD_P      | "P": Start cyclic A/D-conversion at channel Pressure P                                                         | 50µs<br>+<br>AD-<br>conversion<br>time                           |
| D1                                                                                                                                    |        | START_AD_T1     | "T1": Start cyclic A/D-conversion at channel Temperature T1                                                    |                                                                  |
| D2                                                                                                                                    |        | START_AD_T2     | "T2": Start cyclic A/D-conversion at channel Temperature T2                                                    |                                                                  |
| D4                                                                                                                                    |        | START_AD_PAZ    | "PAZ": Start cyclic A/D-conv at channel Auto-Zero Pressure PAZ                                                 |                                                                  |
| D5                                                                                                                                    |        | START_AD_TAZ1   | "T1AZ": Start cyclic A/D-conv at channel Auto-Zero Temp TAZ1                                                   |                                                                  |
| D6                                                                                                                                    |        | START_AD_TAZ2   | "T2AZ": Start cyclic A/D-conv at channel Auto-Zero TempTAZ2                                                    |                                                                  |
| D8                                                                                                                                    |        | START_AD_P_AZ   | "P_AZC": Start cyclic A/D-conversion at channel Pressure P incl.<br>Auto-Zero-Correction                       |                                                                  |
| D9                                                                                                                                    |        | START_AD_T1_AZ  | "T1_AZC": Start cyclic A/D-conversion at channel Temperature T1<br>incl. Auto-Zero-Correction                  |                                                                  |
| DA                                                                                                                                    |        | START_AD_T2_AZ  | "T2_AZC": Start cyclic A/D-conversion at channel Temperature T2<br>incl. Auto-Zero-Correction                  |                                                                  |
| DB                                                                                                                                    |        | START_AD_CMV_AZ | "CMV_AZC": Start cyclic A/D-conversion at channel Common Mode<br>Voltage CMV incl. Auto-Zero-Correction (T1AZ) |                                                                  |

**Table 9:** Additional command set for serial digital interface

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**Note:** These commands are only available in CM.  
CM is set by command START\_CM (0x72<sub>hex</sub>) as first command after power on.

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### 5. EEPROM & RAM

#### 5.1 Programming the EEPROM

Programming the EEPROM is done using an internal charge pump to generate the required programming voltage. The timing of the programming pulses is controlled internally. The programming time for a write-operation is 12.5ms on the condition that the system clock is adjusted. The programming time varies for each IC due to the tolerances of the system clock frequency. Thus a programming time of 20 ms for a write-operation is long enough to program all ICs definitely.

The programming of the EEPROM is done via serial digital interface by sending certain commands (refer chapter 4.5). The commands WRITE\_EEP0 / WRITE\_EEP1 include the address of the EEPROM-word and are followed by the two data bytes. There is also the command CLEAR\_EEP that sets the whole EEPROM to zero. During programming the EEPROM the serial digital interface is disabled so that no commands can be received. In I<sup>2</sup>C mode no acknowledges would be generated from slave ZMD31050.

The additional command COPY\_RAM2EEP writes the contents of the RAM area where EEPROM contents are mirrored back to the EEPROM. This is to simplify the calibration process when the ZMD31050 is configured iteratively. This copy-operation includes 32 write-operations and requires therefore typically 400ms.

Because programming the EEPROM (write, clear, copy RAM to EEPROM) changes the configuration of the device this action is only allowed in CM (start with START\_CM as first command after power on).

#### 5.2 EEPROM & RAM Contents

The configuration of the ZMD31050 is determined by EEPROM programming. There are 22 calibration constants for conditioning the sensor signal by conditioning calculation, 7 configuration words for setting the configuration regarding the application, 1 CRC for checking the validity of EEPROM contents and 2 additional 16bit words for arbitrary free user data.

After every power-on the EEPROM contents are once mirrored to the RAM to avoid frequent access to EEPROM during normal operation. During this read-out and only then the CRC is checked. If EEPROM contents are correct, the sum of all EEPROM-words including the CRC and excluding the free user data is 0xFFFF (calculation with overflow). If a CRC-error is detected the ZMD31050 starts in its reset configuration, opens the time-slot for one wire communication (ZACwire<sup>TM</sup> mode) and finally set the digital serial interface open for I2C and SPI (Open mode).

The configuration of the device is done from mirrored memory area in RAM. Therefore the configuration words are finally transferred to the internal registers. The calibration constants are used for digital conditioning calculation and are also read from RAM. So every change to this RAM area effects the configuration and behavior of the device. The contents of the RAM are after power on determined by EEPROM contents and can then be changed by certain commands writing to RAM. This changed configuration is activated by the command START\_CYC\_RAM.

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| Register-Nr/<br>RAM-Addr<br>hex/dec                                                | Write Command<br>RAM/EEPROM | Description                                                                                 |
|------------------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------|
| <b>Conditioning coefficients - Correction formula pressure (chapter 2.2)</b>       |                             |                                                                                             |
| 0                                                                                  | 80/A0                       | c <sub>0</sub> - Offset                                                                     |
| 1                                                                                  | 81/A1                       | c <sub>1</sub> - Gain                                                                       |
| 2                                                                                  | 82/A2                       | c <sub>2</sub> - Non-linearity 2nd order                                                    |
| 3                                                                                  | 83/A3                       | c <sub>3</sub> - Non-linearity 3rd order                                                    |
| 4                                                                                  | 84/A4                       | c <sub>4</sub> - Temperature coefficient Bridge Offset 1 <sup>st</sup> order                |
| 5                                                                                  | 85/A5                       | c <sub>5</sub> - Temperature coefficient Bridge Offset 2 <sup>nd</sup> order                |
| 6                                                                                  | 86/A6                       | c <sub>6</sub> - Temperature coefficient Gain 1 <sup>st</sup> order                         |
| 7                                                                                  | 87/A7                       | c <sub>7</sub> - Temperature coefficient Gain 2 <sup>nd</sup> order                         |
| <b>Conditioning coefficients – Limit the Output at pin OUT (chapter 2.4)</b>       |                             |                                                                                             |
| 8                                                                                  | 88/A8                       | I <sub>min</sub> - Lower Limit with reference to pin OUT for voltage, current or PWM output |
| 9                                                                                  | 89/A9                       | I <sub>max</sub> - Upper Limit with reference to pin OUT for voltage, current or PWM output |
| <b>Conditioning coefficients - Correction formula temperature (chapter 2.3)</b>    |                             |                                                                                             |
| A / 10                                                                             | 8A/AA                       | t <sub>10</sub> - Offset Temperature 1                                                      |
| B / 11                                                                             | 8B/AB                       | t <sub>11</sub> - Gain Temperature 1                                                        |
| C / 12                                                                             | 8C/AC                       | t <sub>12</sub> - Non-linearity 2 <sup>nd</sup> order Temperature 1                         |
| D / 13                                                                             | 8D/AD                       | t <sub>20</sub> - Offset Temperature 2                                                      |
| E / 14                                                                             | 8E/AE                       | t <sub>21</sub> - Gain Temperature 2                                                        |
| F / 15                                                                             | 8F/AF                       | t <sub>22</sub> - Non-linearity 2 <sup>nd</sup> order Temperature 2                         |
| <b>Conditioning coefficients - Limit the Output at pins IO1, IO2 (chapter 2.4)</b> |                             |                                                                                             |
| 10 / 16                                                                            | 90/B0                       | Alarm 1 : Threshold                      PWM1 : Lower Limit                                 |
| 11 / 17                                                                            | 91/B1                       | Alarm 1 : Hysteresis                      PWM1 : Upper Limit                                |
| 12 / 18                                                                            | 92/B2                       | Alarm 1 : On-/Off-delay (two 8 bit values)                                                  |
| 13 / 19                                                                            | 93/B3                       | Alarm 2 : Threshold                      Common-mode voltage : Lower Limit                  |
| 14 / 20                                                                            | 94/B4                       | Alarm 2 : Hysteresis                      Common-mode voltage : Upper Limit                 |
| 15 / 21                                                                            | 95/B5                       | Alarm 2 : On-/Off-delay (two 8 bit values)                                                  |
| <b>Configuration words (chapter 5.3)</b>                                           |                             |                                                                                             |
| 16 / 22                                                                            | 96/B6                       | CFGCYC: Configuration of measurement cycle                                                  |
| 17 / 23                                                                            | 97/B7                       | CFGSIF: Configuration of digital serial interface                                           |
| 18 / 24                                                                            | 98/B8                       | CFGAPP: Configuration of target application                                                 |
| 19 / 25                                                                            | 99/B9                       | CFGAFE: Configuration of analog front end                                                   |
| 1A / 26                                                                            | 9A/BA                       | CFGTMP: Configuration of temperature measurement                                            |
| 1B / 27                                                                            | 9B/BB                       | CFGOUT: Configuration of signal outputs                                                     |
| 1C / 28                                                                            | 9C/BC                       | ADJREF: Adjustment of internal references                                                   |
| <b>CRC</b>                                                                         |                             |                                                                                             |
| 1D / 29                                                                            | 9D/BD                       | Signature                                                                                   |
| <b>User free memory</b>                                                            |                             |                                                                                             |
| 1E / 30                                                                            | 9E/BE                       | Free user memory, not included in Signature                                                 |
| 1F / 31                                                                            | 9F/BF                       | Free user memory, not included in Signature                                                 |

**Table 10:** EEPROM & RAM contents

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### 5.3 EEPROM & RAM Configuration words

The configuration words are stored in RAM & EEPROM at addresses 22 to 28 (16<sub>hex</sub> to 1C<sub>hex</sub>) and determine the various modes of the ZMD31050. In the following tables the meaning of these configuration words is explained.

#### RAM/EEPROM address 22 / 16<sub>hex</sub>

| Bit     | IC-default | CFGCYC - Configuration of measurement cycle                                                                                                                                                                         |        |
|---------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 15 : 10 | 0000 00    | <i>Not used</i>                                                                                                                                                                                                     | -      |
| 9 : 7   | 00 0       | Count of pressure measurements per cycle<br>Count <i>npmc</i> of pressure measurements between two special measurements (e.g. temperature, auto-zero measurement)<br>$npmc = 2^{PMC} = 1, 2, 4, 8, 16, 32, 64, 128$ | PMC    |
| 6 : 5   | 10         | Access port IO1 (output slot 2)<br>Output of result via port IO1 for:<br>0d Measurand<br>10 Temperature 1<br>11 Temperature 2<br>Active only if CFGOUT:PMIO1 = 10 (PWM output at port IO1).                         | ACOS2  |
| 4 : 3   | 0 1        | Access port OUT (output slot 1)<br>Output of result via port OUT for:<br>0d Measurand<br>10 Temperature 1<br>11 Temperature 2                                                                                       | ACOS1  |
| 2       | 0          | Enable common-mode voltage measurement and check<br>Measures common-mode voltage from bridge (pins INP and INN are shorted) and checks if the result is within the limits defined in EEPROM(addr:19-20)             | CMVE   |
| 1       | 0          | Enable temperature 2 measurement<br>Includes the measurement of temperature 2 during measurement cycle.<br>0 disabled<br>1 enabled                                                                                  | T2E    |
| 0       | 0          | Enable Start-up ROM check<br>ROM check is done after Power-on.<br>Start-up time until first valid output increases by 10ms.<br>0 disabled 1 enabled                                                                 | ROMCHE |

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RAM/EEPROM address 23 / 17<sub>hex</sub>

| Bit    | IC-default | CFGSIF - Configuration of digital serial interface (SIF)                                                                                                                                                                                                                            |          |
|--------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 15 : 9 | 0000 000   | <p>Alternative slave address SIF</p> <p>Additional slave address for communication via digital serial interface. The general slave address is 0x78. If enabled both addresses are valid.</p> <p>Valid for I<sup>2</sup>C, SPI and one wire communication if CFGSIF:SIFID2E = 1.</p> | SIFID2   |
| 8      | 0          | <p>Enable one wire communication during measurement cycle</p> <p>Active only if CFGCYC:OWIWIND = 0 (enables start window for one wire communication).</p> <p>Makes possible the read-out of the conditioning results for pressure and temperature via one wire communication.</p>   | OWIE     |
| 7      | 0          | <p>Disable start window for one wire communication</p> <p>Suppresses the 20ms time-slot for starting one wire communication after power-on and after restart the measurement cycle by sending the commands START_CYC_EEP or START_CYC_RAM</p>                                       | OWIWIND  |
| 6      | 0          | <p>Enable SIF output of conditioning result temperature 2</p> <p>Delivers a continuously updated 15 bit value of temperature 2 for read out during measurement cycle via digital serial interface.</p>                                                                              | SIFOUTT2 |
| 5      | 1          | <p>Enable SIF output of conditioning result temperature 1</p> <p>Delivers a continuously updated 15 bit value of temperature 1 for read out during measurement cycle via digital serial interface.</p>                                                                              | SIFOUTT1 |
| 4      | 1          | <p>Enable SIF output of conditioning result pressure</p> <p>Delivers a continuously updated 15 bit value of measurand for read out during measurement cycle via digital serial interface.</p>                                                                                       | SIFOUTP  |
| 3      | 0          | <p>Enable alternative slave address SIF</p> <p>The alternative slave address CFGSIF:SIFID2 is valid for I<sup>2</sup>C, SPI and one wire communication.</p>                                                                                                                         | SIFID2E  |
| 2      | 1          | <p>SPI clock edge select</p> <p>0 sample leading edge      1 sample trailing edge</p>                                                                                                                                                                                               | SPICKE   |
| 1      | 0          | <p>SPI clock polarity</p> <p>0 SCK low, idle      1 SCK high, idle</p>                                                                                                                                                                                                              | SPICKP   |
| 0      | 0          | <p>SIF mode</p> <p>Chose active communication mode:</p> <p>0 I<sup>2</sup>C communication</p> <p>1 SPI communication</p>                                                                                                                                                            | SIFMD    |

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RAM/EEPROM address 24 / 18<sub>hex</sub>

| Bit     | IC-default | CFGAPP - Configuration of target application                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |
|---------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 15      | 0          | Bridge Signal Polarity - Interchange of input signal (pin VINP and VINN)<br>0 positive ( $V_{BR\_DIFF} = V_{BR\_P} - V_{BR\_N}$ )<br>1 negative ( $V_{BR\_DIFF} = V_{BR\_N} - V_{BR\_P}$ )                                                                                                                                                                                                                                                                                              | BSP     |
| 14      | 1          | Not used – Set always to '1'                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| 13 : 12 | 11         | ADC Range Shift regarding signal from IN3 (temperature 2)<br>Active only if CFGTMP:TAM2 = 11 (Measurement of T2 with arbitrary voltage via pin IN3).<br>00 ADC Input Range = $[(-1/16 V_{ADC\_REF}) \text{ to } (+15/16 V_{ADC\_REF})]$<br>01 ADC Input Range = $[(-1/8 V_{ADC\_REF}) \text{ to } (+7/8 V_{ADC\_REF})]$<br>10 ADC Input Range = $[(-1/4 V_{ADC\_REF}) \text{ to } (+3/4 V_{ADC\_REF})]$<br>11 ADC Input Range = $[(-1/2 V_{ADC\_REF}) \text{ to } (+1/2 V_{ADC\_REF})]$ | ADRAIN3 |
| 11 : 10 | 01         | IN3 reference mode<br>Active only if CFGTMP:TAM2 = 11 (Measurement of T2 with arbitrary voltage via pin IN3). Arbitrary voltage is measured referenced to:<br>00 Bandgap reference $V_{BG}$ (approx. 1.23V)<br>01 Supply voltage $V_{DDA} / 2$<br>1d Supply voltage check (Measures $V_{DDA} / 2$ referenced to $V_{BG}$ )                                                                                                                                                              | IN3M    |
| 9       | 1          | Activate sensor connection and RAM (=config registers) parity check:<br>0 enabled 1 disabled                                                                                                                                                                                                                                                                                                                                                                                            | SCCD    |
| 8       | 0          | Sensor bridge excitation mode :<br>0 Voltage supplied<br>1 Current supplied (constant current generated with internal regulator and external low-TC resistor)<br><b>hint: CFGAPP:XZCE=1 is required for CFGAPP:CSBE=1</b>                                                                                                                                                                                                                                                               | CSBE    |
| 7       | 0          | Enable eXtended Zero Compensation<br>offset compensation by analog front end, also called AnalogZeroPointShift, value is defined with CFGAFE:XZC<br>0 disabled 1 enabled                                                                                                                                                                                                                                                                                                                | XZCE    |
| 6 : 5   | 01         | Raw oscillator frequency adjust<br>Fine adjustment by ADJREF:OSCA<br>00 $f_{CLK} = 1 \text{ MHz}$ 10 $f_{CLK} = 2 \text{ MHz}$<br>01 $f_{CLK} = 2 \text{ MHz}$ 11 $f_{CLK} = 4 \text{ MHz}$                                                                                                                                                                                                                                                                                             | OSCF    |
| 4       | 0          | Enable external clock<br>0 disabled 1 enabled                                                                                                                                                                                                                                                                                                                                                                                                                                           | ECLKE   |
| 3       | 0          | AD-Converter reference voltage selection<br>0 ADC-Ref = $V_{BR}$ 1 ADC-Ref = $V_{DDA}$<br><b>hint: CFGAPP:XZCE=1 &amp; CFGAFE:XZC=0 is required for CFGAPP:ADREF=1 &amp; CFGAPP:CSBE=1</b>                                                                                                                                                                                                                                                                                              | ADREF   |
|         |            | ... continued on next page                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |

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Functional Description

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### RAM/EEPROM address 24 / 18<sub>hex</sub>

| Bit   | IC-default | CFGAPP - Configuration of target application                                                                                                               |      |
|-------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 2 : 1 | 10         | Value of controlled analog supply voltage.<br>00 VDDA = 3.0 V                      10 VDDA = 5.0 V<br>01 VDDA = 4.0 V                      11 VDDA = 5.5 V | VDC  |
| 0     | 1          | Enable analog supply voltage control.<br>Uses internal controller with external transistor.                                                                | VDCE |

### RAM/EEPROM address 25 / 19<sub>hex</sub>

| Bit     | IC-default | CFGAFE - Configuration of analog front end                                                                                                                                                                                                                                                                                                                                                                                                             |        |
|---------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 15 : 13 | 100        | Resolution of A/D-conversion ( $r_{ADC}$ , see chapter 2.1)<br>Valid for pressure as well as for temperature measurement.<br>Influences conversion time and integration time of measurement.<br>000 9 Bit                                      100 13 Bit<br>001 10 Bit                                     101 14 Bit<br>010 11 Bit                                     11d 15 Bit<br>011 12 Bit                                                      | RADC   |
| 12      | 1          | Order of A/D-conversion<br>Influences conversion time and integration time of measurement.<br>0 1 <sup>st</sup> order conversion<br>1 2 <sup>nd</sup> order conversion                                                                                                                                                                                                                                                                                 | OADC   |
| 11 : 10 | 10         | ADC Range Shift regarding measurand signal<br>( $RS_{ADC}$ , see chapter 2.1)<br>00 ADC Input Range = $[(-1/16 V_{ADC\_REF}) \text{ to } (+15/16 V_{ADC\_REF})]$<br>01 ADC Input Range = $[(-1/8 V_{ADC\_REF}) \text{ to } (+7/8 V_{ADC\_REF})]$<br>10 ADC Input Range = $[(-1/4 V_{ADC\_REF}) \text{ to } (+3/4 V_{ADC\_REF})]$<br>11 ADC Input Range = $[(-1/2 V_{ADC\_REF}) \text{ to } (+1/2 V_{ADC\_REF})]$                                       | ADRAPR |
| 9 : 4   | 00 0000    | Value of eXtendedZeroCompensation (offset compensation by analog front end $AZS_{IN}$ ; refer chapter 2.1 and DS chapter 1.3)<br>Active only if CFGAPP:XZCE = 1. The width/value of a extended zero compensation step depends on the chosen input span. Refer datasheet chapter 1.3 for correlation between XZC adjustment and resulting offset cancellation.                                                                                          | XZC    |
| 3 : 0   | 0000       | Gain analog front end measurand measurement ( $a_{IN\_p}$ , refer chapter 2.1)<br>0000 420                                      0111 35<br>0001 280                                      1000 26.25<br>0010 210                                      1001 14<br>0011 140                                      1010 9.3<br>0100 105                                      1011 7<br>0101 70                                        11dd 2.8<br>0110 52.5 | GAIN   |

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RAM/EEPROM address 26 / 1A<sub>hex</sub>

| Bit     | IC-default | CFGTMP - Configuration of temperature measurement                                                                                                                                                                                                                                                                                     |        |
|---------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 15 : 14 | 00         | Gain analog front end temperature 2 meas. (a <sub>IN_T2</sub> , see chapter 2.1)<br>00: GT4 (≈ 5,65)                      10: GT2 (≈ 4,7)<br>01: GT3 (≈ 5,17)                      11: GT1 (≈ 1,9)<br>For details see chapter 6.                                                                                                      | GAINT2 |
| 13 : 12 | 00         | Gain analog front end temperature 1 meas. (a <sub>IN_T1</sub> , see chapter 2.1)<br>00: GT4 (≈ 5,65)                      10: GT2 (≈ 4,7)<br>01: GT3 (≈ 5,17)                      11: GT1 (≈ 1,9)<br>For details see chapter 6.                                                                                                      | GAINT1 |
| 11      | 0          | Polarity of external temperature sensor for temperature 2 meas.<br>Active only if CFGAPP:TAD2 = 0 and CFGTMP:TAM2 ≠ 11<br>(enables temperature 2 measurement via pin TEMP).<br>Switches polarity of supply current for temperature sensor.<br>0    Temperature sensor to VSS<br>1    Temperature sensor to VDDA                       | PETS2  |
| 10      | 0          | Polarity of external temperature sensor for temperature 1 meas.<br>Switches polarity of supply current for temperature sensor.<br>0    Temperature sensor to VSS<br>1    Temperature sensor to VDDA                                                                                                                                   | PETS1  |
| 9 : 7   | 010        | Temperature 2 (T2) zero point adjust<br>Allows shift of characteristics of temperature measurement.<br>Refer chapter 6 for details.                                                                                                                                                                                                   | ZCT2   |
| 6 : 4   | 010        | Temperature 1 (T1) zero point adjust<br>Allows shift of characteristics of temperature measurement.<br>Refer chapter 6 for details.                                                                                                                                                                                                   | ZCT1   |
| 3 : 2   | 01         | Temperature 2 (T2) acquisition mode<br>Defines the used temperature sensor connected to pin IR_TEMP<br>and implicit the input voltage range of temperature signal.<br>00   Internal diode                      01    External diode<br>10   External resistor                  11    External voltage<br>Refer chapter 6 for details. | TAM2   |
| 1 : 0   | 00         | Temperature 1 (T1) acquisition mode<br>Defines the used temperature sensor connected to pin IR_TEMP<br>and implicit the input voltage range of temperature signal.<br>00   Internal diode                      01    External diode<br>1d   External resistor<br>Refer chapter 6 for details.                                         | TAM1   |

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RAM/EEPROM address 27 / 1B<sub>hex</sub>

| Bit     | IC-default | CFGOUT - Configuration of signal outputs                                                                                                                                                                                                                                                                                                                                                                     |         |
|---------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 15 : 14 | 10         | Resolution PWM output<br>Active only if CFGOUT:PMIO1 = 10 or CFGOUT:COS1 = 10.<br>Influences the PWM output frequency.<br>00 9 Bit 10 11 Bit<br>01 10 Bit 11 12 Bit                                                                                                                                                                                                                                          | PWMRES  |
| 13 : 12 | 00         | Clock divider PWM output regarding internal oscillator frequency<br>Influences the PWM output frequency (see Table 4).<br>00 f <sub>CLK</sub> / 1 10 f <sub>CLK</sub> / 4<br>01 f <sub>CLK</sub> / 2 11 f <sub>CLK</sub> / 8                                                                                                                                                                                 | PWMD    |
| 11 : 10 | 00         | Output polarity { PWM2 / PWM1 } at pin { OUT / IO1 }<br>0 PWM period starts with high level (high-active)<br>1 PWM period starts with low level (low active)                                                                                                                                                                                                                                                 | PWMPO   |
| 9       | 0          | No access                                                                                                                                                                                                                                                                                                                                                                                                    |         |
| 8       | 0          | Clock divider for A/D-converter<br>Influences only the frequency for the A/D-converter. Does not influence the internal working frequency.<br>Set in connection with f <sub>CLK</sub> = 4MHz (CFGAPP:OSCF = 11) to enlarge the processing time of internal micro-controller when low resolution of less or equal then 12 Bit (CFGAFE:RADC = 0dd) is chosen.<br>0 f <sub>CLK</sub> / 1 1 f <sub>CLK</sub> / 2 | VFCCLKD |
| 7       | 0          | Enable Alarm window mode<br>Both thresholds and hysteresis (EEPROM addresses 16/10 <sub>hex</sub> , 17/11 <sub>hex</sub> , 19/13 <sub>hex</sub> , 20/14 <sub>hex</sub> ) are used to define a window for alarm output. Alarm 2 at pin IO2 is inverted to Alarm 1 at pin IO1.<br>0 disabled 1 enabled                                                                                                         | AWME    |
| 6 : 5   | 11         | Output polarity { Alarm 2 / Alarm 1 } at pin { IO2 / IO1 }<br>0 Alarm low-active 1 Alarm high-active                                                                                                                                                                                                                                                                                                         | APO     |
| 4       | 0          | Output mode port IO2<br>Threshold, hysteresis and delay of Alarm 2 output are defined in EEPROM addresses 19/13 <sub>hex</sub> to 21/15 <sub>hex</sub> .<br>0 disabled 1 Alarm 2 output                                                                                                                                                                                                                      | PMIO2   |
| 3 : 2   | 00         | Output mode port IO1<br>Threshold, hysteresis and delay of Alarm 1 output respectively the limits for PWM1 output are defined in EEPROM addresses 16/10 <sub>hex</sub> to 18/12 <sub>hex</sub> .<br>0d Disable 10 PWM1 output 11 Alarm 1 output                                                                                                                                                              | PMIO1   |
| 1 : 0   | 00         | Output mode port OUT<br>The limits for all outputs are defined in EEPROM addresses 8 to 9.<br>00 Voltage output 01 Current output<br>10 PWM2 output 11 Disable                                                                                                                                                                                                                                               | COS1    |

RAM/EEPROM address 28 / 1C<sub>hex</sub>

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| Bit     | IC-default | ADJREF - Adjustment of internal references                                                                                                                                                                                                                                                                                                                                                                                                   |      |
|---------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 15 : 13 | 100        | Adjust sensor bridge current<br>Active only if CFGAPP:CSBE = 1 (enables current supply for sensor bridge). Supply current depends on external zero-TC reference resistor $R_{BR\_REF}$ $I_{BR} = VDDA / (16 \cdot R_{BR\_REF})$ .<br>$I_{BR}$ is fine adjustable in the range of 0,5 to $1,375 \cdot I_{BR}$ . Default setup is equivalent factor 1 and one adjustment step increments (->111) /decrements (->000) factor about 0.125 units. | CSB  |
| 12 : 10 | 100        | Fine adjustment of controlled analog supply voltage<br>The controlled analog supply voltage is enabled with CFGAPP:VDCE and is raw adjusted with CFGAPP:VDC.<br>Use this value to fine adjust the controlled analog supply voltage.                                                                                                                                                                                                          | VDCA |
| 9 : 7   | 100        | Adjust oscillator frequency<br>Internal oscillator frequency $f_{CLK}$ is raw adjusted with CFGAPP:OSCF. Use this value to fine adjust the internal oscillator frequency in the range of 0,8 to $1,15 \cdot f_{CLK\_NOM}$                                                                                                                                                                                                                    | OSCA |
| 6 : 4   | 100        | Adjust bias current<br>Internal bias current is adjustable to move a total current consumption in required range. Current consumption can be decreased/increased by this value but may cause worse/better analog performance.                                                                                                                                                                                                                | BCUR |
| 3 : 0   | 1000       | TC adjustment of internal bandgap voltage<br>Used to adjust temperature behaviour of the controlled analog supply voltage.<br>Hint: The controlled analog supply voltage is enabled with CFGAPP:VDCE and is raw adjusted with CFGAPP:VDC.                                                                                                                                                                                                    | VREF |

### 5.4 EEPROM Signature

The EEPROM signature (address 29/1D<sub>hex</sub>) is used to check the validity of EEPROM contents. The signature is built using a polynomial arithmetic modulo 2. The following source-code generates the signature if the field eepcont is allocated by the EEPROM contents (address 0 to 28) and N = 29 is the count of considered addresses.

```
#define POLYNOM 0xA005
unsigned short signature(eepcont, N)
unsigned short eepcont[], N;
{
    unsigned short sign, poly, p, x, i, j;
    sign = 0;
    poly = POLYNOM;
    for (i=0; i<N; i++) {
        sign ^= eepcont[i];
        p = 0; x = sign & poly;
        for (j=0; j<16; j++, p ^= x, x >>= 1);
        sign <<= 1; sign += (p & 1);
    }
    return(~sign);
}
```

**Fig. 17:** Source-code signature generation

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## Advanced Differential Sensor Signal Conditioner

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## 6. TEMPERATURE SENSOR ADAPTION & CMV MEASSUREMENT

### 6.1 Sensor Bridge in Voltage Mode

Keep in mind, only T1 can be used for calibration. There are a few configurations for the temperature measurement that make sense for voltage supplied bridges. Table 11 shows these configurations for different types of temperature sensors. Table 12 ff. defines the resulting input range for the differential voltage  $V_{IN\_T}$  at using an external temperature measuring diode for different supply voltages.

| Temperature sensor | Analog Gain $a_{IN\_T}$ | Zero point adjust ZCT | Sensor referenced to                         | Remarks                                                                                      |
|--------------------|-------------------------|-----------------------|----------------------------------------------|----------------------------------------------------------------------------------------------|
| Internal diode     | GT2                     | 5                     | VSS                                          | recommended adjust                                                                           |
| External diode     | GT4 / GT3 / GT2         | 0 ... 7               | VSS $V_{IN\_DIFF\_T} = V_{IR\_TEMP} - VSS$   | $V_{IR\_TEMP}$ is voltage at pin IR_TEMP;<br>$V_{IN\_DIFF\_T}$ is differential input voltage |
| External resistor  | GT4 / GT3 / GT2         |                       | VDDA $V_{IN\_DIFF\_T} = VDDA - V_{IR\_TEMP}$ |                                                                                              |

**Table 11:** Configuration of temperature measurement

#### 6.1.1 Internal and External Diode

Following the adapting of an external diode is described. The red marked fields are recommended values for typical diodes in temperature range. Measure  $V_{IN\_DIFF\_T}$  to find an adjustment, normally 650mV are expected. Typically  $V_{IN\_DIFF\_T}$  is changing depending on temperature with  $-2mV/K$ .

**Hint:** At using an internal or external diode for temperature measurement it is necessary to adapt the supply voltage regulator at the external supply voltage (refer register 18<sub>hex</sub> CFGAPP:VDC). The voltage regulator has to be tuned to the nearest smaller value than the VDDA-potential. The input voltage ranges are valid for the described adjustments. At adapting an external temperature measuring diode has to be secured, that at minimum and maximum temperature the  $V_F$  of the diode including tolerances is inside the described voltage range.

**The temperature sensor has to be adjusted in gain and offset shift in such manner, that the measurement readout (command: "D9" - START\_AD\_T1) in full temperature range is inside of 12.5-87.5% of ADC range (e.g. 13bit, +/- 1/2: ADC-min/max = -3072/3072). Described input signal range has to be noticed add on.**

| CFGAPP:VDDA=5V ( $V_{T\_REF}=4V$ ) |                   | Tx ZeroPointShift ZCT |      |      |      |      |      |      |      |
|------------------------------------|-------------------|-----------------------|------|------|------|------|------|------|------|
| Analog Gain $a_{IN\_T}$            | $V_{IN\_DIFF\_T}$ | 0                     | 1    | 2    | 3    | 4    | 5    | 6    | 7    |
| GT4                                | min [V]           | 0,12                  | 0,24 | 0,35 | 0,46 | 0,97 | 1,01 | 1,12 | 1,15 |
|                                    | max [V]           | 0,6                   | 0,71 | 0,82 | 0,9  | 1,56 | 1,64 | 1,74 | 1,77 |
| GT3                                | min [V]           | External Diode        |      |      | 0,25 | 0,5  |      |      |      |
|                                    | max [V]           |                       |      |      | 0,60 | 1,0  |      |      |      |
| GT2                                | min [V]           |                       |      |      | 0,05 | 0,3  | 0,35 | 0,39 |      |
|                                    | max [V]           |                       |      |      | 0,6  | 0,75 | 0,8  | 0,85 |      |

**Table 12:** ZMD31050C VDDA=5V: Input signal range  $V_{IN\_DIFF\_T}$  at using an external diode (to VSS)

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| CFGAPP:VDDA=4V (V <sub>T_REF</sub> =3,25V)           |         | Tx ZeroPointShift ZCT |   |      |      |      |                |   |   |
|------------------------------------------------------|---------|-----------------------|---|------|------|------|----------------|---|---|
| Analog Gain a <sub>IN_T</sub> V <sub>IN_DIFF_T</sub> |         | 0                     | 1 | 2    | 3    | 4    | 5              | 6 | 7 |
| GT4                                                  | min [V] |                       |   | 0,28 | 0,37 |      | External Diode |   |   |
|                                                      | max [V] |                       |   | 0,54 | 0,63 |      |                |   |   |
| GT3                                                  | min [V] |                       |   | 0,02 | 0,08 | 0,37 | 0,43           |   |   |
|                                                      | max [V] |                       |   | 0,45 | 0,51 | 0,8  | 0,9            |   |   |
| GT2                                                  | min [V] |                       |   |      |      |      |                |   |   |
|                                                      | max [V] |                       |   |      |      |      |                |   |   |

**Table 13: ZMD31050C VDDA=4V: Input signal range  $V_{IN\_DIFF\_T}$  at using an external diode (to VSS)**

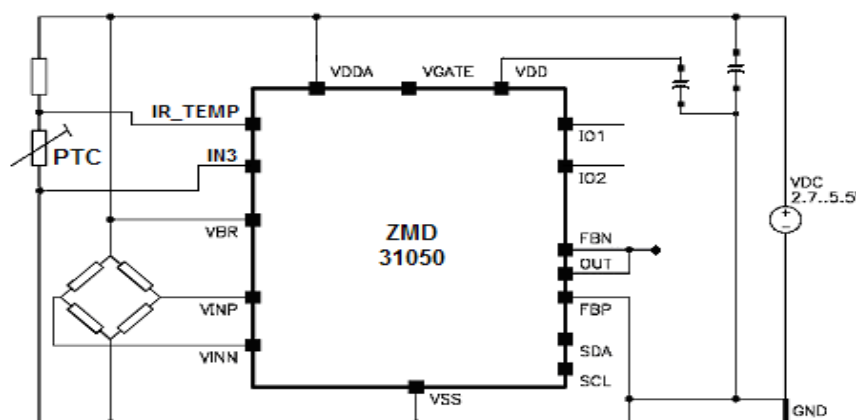
| CFGAPP:VDDA=3V ( $V_{T\_REF}=2,5V$ ) |                   | Tx ZeroPointShift ZCT |   |   |      |      |      |      |      |
|--------------------------------------|-------------------|-----------------------|---|---|------|------|------|------|------|
| Analog Gain $a_{IN\_T}$              | $V_{IN\_DIFF\_T}$ | 0                     | 1 | 2 | 3    | 4    | 5    | 6    | 7    |
| GT4                                  | min [V]           | External Diode        |   |   | 0,28 | 0,62 |      |      |      |
|                                      | max [V]           |                       |   |   | 0,53 | 0,87 |      |      |      |
| GT3                                  | min [V]           |                       |   |   | 0,06 | 0,30 | 0,35 | 0,40 | 0,44 |
|                                      | max [V]           |                       |   |   | 0,43 | 0,64 | 0,70 | 0,75 | 0,80 |
| GT2                                  | min [V]           |                       |   |   |      |      |      |      |      |
|                                      | max [V]           |                       |   |   |      |      |      |      |      |

**Table 14: ZMD31050C VDDA=3V: Input signal range  $V_{IN\_DIFF\_T}$  at using an external diode (to VSS)**

**Hint:** The input signal ranges  $V_{IN\_DIFF\_T}$  are roughly estimated and have to verify in application.

### 6.1.2 External Resistor

The external resistor mode supports using an external half bridge for the temperature measurement, which is connected between VDDA and VSS. The input signal range is asymmetric and ends in maximum approx.  $30\% \cdot VDDA$  less than VDDA or  $30\% \cdot VDDA$  more than VSS (programmable reference VDDA or VSS). Table 15 explains resulting input range for using an external resistor for temperature measurement in detail. The voltage  $V_{IN\_DIFF\_T}$  is displayed as a ratio to VDDA because temperature measurement via an external resistor delivers a ratiometric result.



**Fig. 18: Temperature measurement with external resistor (PTC) (input range asymmetric,  $0 \dots 30\% \cdot VDDA$ , beginning from VSS)**

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|                         |                             | Tx ZeroPointShift ZCT |       |       |       |                   |       |       |       |
|-------------------------|-----------------------------|-----------------------|-------|-------|-------|-------------------|-------|-------|-------|
| Analog Gain $a_{IN\_T}$ | $V_{IN\_DIFF\_T} / V_{DDA}$ | 0                     | 1     | 2     | 3     | 4                 | 5     | 6     | 7     |
| GT4                     | min                         | 3,7%                  | 6,4%  | 9,1%  | 11,8% | External Resistor |       |       |       |
|                         | max                         | 18,0%                 | 20,7% | 23,4% | 26,1% |                   |       |       |       |
| GT3                     | min                         | 0,0%                  | 1,2%  | 3,0%  | 4,8%  | 13,9%             | 15,6% | 17,4% | 19,2% |
|                         | max                         | 15,1%                 | 16,9% | 18,7% | 20,5% | 29,4%             | 31,2% | 33,0% | 34,8% |
| GT2                     | min                         |                       |       | 0,0%  | 0,9%  | 7,6%              | 9,0%  | 10,3% | 11,6% |
|                         | max                         |                       |       | 16,6% | 18,1% | 24,7%             | 26,0% | 27,4% | 28,7% |

**Table 15:** ZMD31050C VDDA = 3 ... 5V: Valid input signal range ( $V_{IN\_DIFF\_T} / V_{DDA}$ ) using external resistor mode

Hint: The input signal ranges ( $V_{IN\_DIFF\_T} / V_{DDA}$ ) are roughly estimated and have to verify in application.

### 6.1.3 Conversion Result and Sensitivity Calculation

The gain (=  $a_{IN\_T}$ ) and offset (= ZCT) adjustments for the temperature measurement are programmable. The gain factor  $a_{IN\_T}$  is a calculated value for a estimated calculation of sensitivity. The temperature measurement result is referenced to  $V_{T\_REF}$  (depending on input mode) and can be calculated and verified by using following formulas (voltages referenced to VSS,  $V_{IR\_TEMP}$  = potential at pin IR\_TEMP referenced to VDDA or VSSA depending on adjust):

| Gain Identifier | $a_{IN\_T}$ | $G_{FB2}$ | $G_{XZC}$ | Gain Ident in old software |
|-----------------|-------------|-----------|-----------|----------------------------|
| GT4             | 5,65        | 6         | 3         | 14                         |
| GT3             | 5,17        | 5,5       | 1,833     | 9                          |
| GT2             | 4,7         | 5         | 1,25      | 7                          |
| GT1             | 1,88        | 2         | 0,2       | 2.8                        |

**Table 16:** gain calculation parameter

$$T1 = \left[ 2 \cdot \left( G_{XZC} \cdot \frac{Z_{XZC}}{40} - \frac{V_{IR\_TEMP}}{V_{T\_REF}} \cdot \frac{G_{FB2}}{2,125} \right) + \frac{1}{2} \right] \cdot 2^{ADCRES} \quad \Rightarrow \text{cmd "D1"}$$

$$T1AZ = \frac{1}{2} \cdot 2^{ADCRES} \quad \Rightarrow \text{cmd "D5"}$$

$$T1\_AZC = T1 - T1AZ = \left( G_{XZC} \cdot \frac{Z_{XZC}}{40} - \frac{V_{IR\_TEMP}}{V_{T\_REF}} \cdot \frac{G_{FB2}}{2,125} \right) \cdot 2^{ADCRES+1} \quad \Rightarrow \text{cmd "D9"}$$

$Z_{XZC} = \{4,5,6,7,12,13,14,15\}$  for software adjust ZeroPointShift ZCT =  $\{0,...,7\}$

$V_{T\_REF} = 4/3.25/2.5$  for diode or VDDA for ext. resistor

ADCRES = ADC-resolution in bit

The resolution of temperature measurement can be calculated and verified by using following formula:

$$ST_{TS\_E} = \text{ABS} \{ [T1\_AZC(V_{IR\_TEMP\_X}) - T1\_AZC(V_{IR\_TEMP\_Y})] / [V_{IR\_TEMP\_X} - V_{IR\_TEMP\_Y}] \} \quad [\text{counts/V}]$$

$$ST_{TS\_E} = 2^{ADCRES} \cdot (a_{IN\_T} / V_{T\_REF}) \quad [\text{counts/V}]$$

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### 6.2 Sensor Bridge in Current Mode

Bridge current excitation enables temperature measurement using temperature coefficient of bridge resistors – so no additional temperature sensor is needed. For temperature data acquiring the common mode voltage of the bridge inputs (VINN & VINP) is measured.

The bridge current has to be adjusted with restriction that the voltage over bridge  $V_{DDBR}$  and the input signals  $V_{BR\_P}$  and  $V_{BR\_N}$  are in the allowed range. The bridge current can be tuned coarse by changing external resistor  $R_{BR\_REF}$  and adjusted fine by changing configuration register  $ADJREF:CSB$ .

The reference resistor  $R_{BR\_REF}$  can be calculated as follows:

$$R_{BR\_REF} = R_{BR}(T) * V_{DDA} / (16 * 2 * V_{IN\_CM})$$

The subsection of  $I_{BR}$  from  $R_{BR\_REF}$  can be explained as follows:

$$I_{BR} = V_{DDA} / (16 * R_{BR\_REF})$$

This can be used to calculate the current:

$$I_{BR} = 2 * V_{IN\_CM} / R_{BR} \text{ (with } I_{BR} < 1.5\text{mA)}$$

$V_{IN\_CM}$  in bridge excitation mode depends on a set of parameters, that's why ZMD recommends to use "ZMD31050\_Bridge\_Current\_Excitation\_Rev\_\*.xls" adjustment calculation sheet for a detailed calculation of  $R_{BR\_REF}$ ,  $R_{TOP}$  and  $R_{BOT}$ . So it is secured, that the common mode range is not loosed under consideration full temperature behaviour, tolerances of external elements and measured rejection limits.

Bridge current excitation mode offers two possibilities for selection of AD converter reference voltage, which is adjusted with  $CFGAPP:ADREF$  configuration bit. In "VREF=VBR" mode the potential of the pin VBR is used as AD converters reference voltage, in "VREF=VDDA" mode VDDA is used as AD converters reference voltage. "VREF=VDDA" has to be used for sensor bridges with nonratiometric behaviour in temperature range like temperature precompensated sensor.

In case of activated bridge current excitation XZCE has to be enabled always, if "VREF=VDDA" is additional active, XZC has to be set to "0". Following rules describes these behaviour:

$$CFGAPP:CSBE=1 \Rightarrow CFGAPP:XZCE=1$$

$$CFGAPP:ADREF=1 \text{ \& } CFGAPP:CSBE=1 \Rightarrow CFGAPP:XZCE=1 \text{ \& } CFGAPP:XZC=0$$

#### 6.2.1 Bridge signal measurements

Calculation schema in "ZMD31050\_Bridge\_Current\_Excitation\_Rev\_\*.xls" verify to abide input signal common mode range depending on gain. The input / common mode voltage ranges depend on adjusted gain  $a_{IN\_2}$  of the second PGA-stage (refer ZMD31050 datasheet, chapter 1.3.1).

For current supplied bridges the temperature measurement T1 is always done by evaluating the common mode voltage of the sensor bridge. The mode is enabled by  $CFGAPP:CSBE = 1$ . The necessary analog gain  $a_{IN\_T} = 2.8$  and has to be set for configuration parameter  $CFGTMP:GAINT1$ . The parameter  $CFGTMP:TAM1$  and  $CFGTMP:ZCT1$  are not considered.

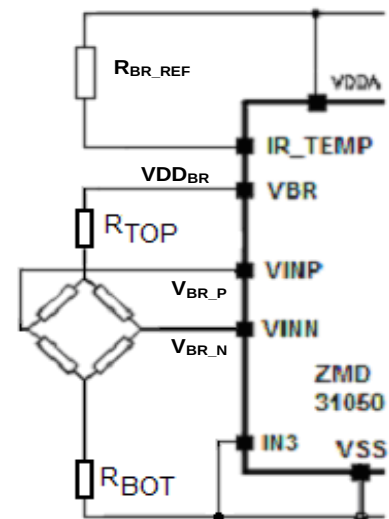


Fig. 19: Bridge current mode application

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It is possible to verify input signal common mode level/potential at calibration using commands "D9" or "DB". Calculation schema "ZMD31050\_Bridge\_Current\_Excitation\_Rev\_\*.xls" represents also ideal AD conversion result /data readout for commands "D9" and "DB". ZMD recommends to use this function to verify abiding common mode range in application and to interpolate to real common mode potential, if measurement/calibration in temperature range is not located at the edges of temperature of application.

### 6.2.2 Temperature Measurement

Temperature measurement is configured 100%ly through activating "Bridge current Mode" – nothing has to be adjusted. Temperature measurement result can be calculated and verified by using following formulas (voltages referenced to VSS,  $r_{ADC}$  = ADC-resolution in bit):

$$V_{IN\_CM} = (V_{BR\_P} + V_{BR\_N}) / 2$$

$$T1 = 2^{r_{ADC}} * \{2.8 * [V_{IN\_CM} / VDDA - 1/3] + 0.5\} \quad \Rightarrow \text{command "D1"}$$

$$T1AZ = 2^{r_{ADC}} * 0.5 \quad \Rightarrow \text{command "D5"}$$

$$T1\_AZC = T1 - T1AZ = 2^{r_{ADC}} * 2.8 * [V_{IN\_CM} / VDDA - 1/3] \quad \Rightarrow \text{command "D9"}$$

Gain and offset in bridge current mode aren't programmable for temperature acquiring. The resolution of temperature measurement can be calculated and verified by using following formula:

$$ST_{TS\_BCM} = \text{ABS} \{ [T1\_AZC(V_{IN\_CM\_X}) - T1\_AZC(V_{IN\_CM\_Y})] / [V_{IN\_CM\_X} - V_{IN\_CM\_Y}] \} \quad [\text{counts/V}]$$

$$ST_{TS\_BCM} = 2^{r_{ADC}} * (2.8 / VDDA) \quad [\text{counts/V}]$$

### 6.3 TEMP2 Measurements

The TEMP2 path is an additional path, which can be calibrated independent from the main path up to second order in the temperature and the signal (offset, gain and nonlinearity). **TEMP2 measurements can't be used for pressure signal temperature calibration.**

ZMD31050 offers 4 modes for the TEMP2-measurements:

- Internal diode mode (CFGTMP:TAM2=00)
- External diode using pin IR\_TEMP (CFGTMP:TAM2=01)
- Resistor mode using pin IR\_TEMP (CFGTMP:TAM2=10)
- External voltage mode using pin IN3 (CFGTMP:TAM2=11)

#### 6.3.1 Internal, External Diode and Resistor Mode

Refer chapter 6.1 for adapting input constraints and the adjustments for internal/external diode and resistor.

#### 6.3.2 External Voltage Mode (also called IN3-measurement)

The external Voltage Mode realizes ratiometric signal measurement of an external half bridge (connected via pin IN3) or a voltage source referenced to an internal voltage divider or source.

### Adjustable Parameters/Configurations

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The **IN3-Mode** (CFGAPP:IN3M) defines an internal half bridge connectivity = reference potential or measurement mode for IN3-access:

- “V->VDDA/2” VDDA/2 is used as reference potential (ratiometric measurement)
- “V->V<sub>BG</sub>” Bandgap voltage (approx. 1.23V) is used as reference potential (non-ratiometric)
- “VDDA/2->V<sub>BG</sub>” Supply voltage measurement mode – divided supply voltage (VDDA) is measured referenced to bandgap voltage

### Additional adjustment parameters

- **ADC-Range Shift IN3** (CFGAPP:ADRAIN3) defines common mode range of input signal and is used to adjust AD converter range to real appearing input signal range.
- **T2 Analog Gain** (CFGTMP:GAINT2) defines gaining for difference signal between IN3 and internal reference potential.

| Mode               |                  | GT1    | GT2    | GT3    | GT4    |
|--------------------|------------------|--------|--------|--------|--------|
| V->VDDA/2          | min [%VDDA]      | 1,80/5 | 2,20/5 | 2,30/5 | 2,40/5 |
|                    | max [%VDDA]      | 3,30/5 | 2,80/5 | 2,70/5 | 2,70/5 |
| V->V <sub>BG</sub> | min [V] @VDDA=5V | 0,75   | -      | -      | -      |
|                    | max [V] @VDDA=5V | 2,05   | -      | -      | -      |

**Table 18:** IN3 Input Signal Range depending on mode and gain

**Hint:** The input signal ranges for IN3 are roughly estimated and have to verify in the application in detail.

## 6.4 CMV Measurement

ZMD31050 offers a special method for sensor aging detection. The common mode voltage (CMV) of sensor bridge is measured and compared with defined limits for these function. Limits are stored in registers 0x13 (lower) and 0x14 (upper). A CMV error is messaged if current CMV measurement result is lower than value stored in register 0x13 or higher than value stored in register 0x14.

Limits can be expressed as percentage to CMV measurement result (command: “DB”) or calculated directly with CMV\_AZC formula. Notice at definition of CMV limits, that CMV measurement result can drift in temperature range depending on temperature behaviour of sensor bridge.

CMV measurement acquires common mode voltage of bridge (VINcm) and can be calculated using following formula approximately:

$$\text{CMV\_AZC} = 2^{\text{RES}} * (2,8 * (\text{VINcm}/\text{VDDA} - 1/3) + 1/16) - \text{T1AZ} + 2^{\text{RES}}/2$$

with:

RES = Resolution in Bit (RES =[9-15])

T1AZ = AZ-Readout of T1 Measurement (“D5”-command)

VINcm = common mode voltage at input (shorted VINP & VINN)

VDDA = VDDA (supply) voltage

In ideal case is T1AZ = RES/2:

$$\text{CMV\_AZC} = 2^{\text{RES}} * (2,8 * (\text{VINcm}/\text{VDDA} - 1/3) + 1/16)$$

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## 7. ADDITIONAL BUILD IN FEATURES

### 7.1 Internal Supply Regulator using an external FET

In non-ratiometric mode the internal supply regulator can be used for controlling the VDDA supply voltage. The selection of the external regulation FET has to be made by checking the drop and the regulation range under temperature conditions, parameter deviation of the external FET and power consumption of the bridge and the ZMD31050.

Recommended FET-types (JFET or N-Depletion) are:

- J112: used at our kits (can be to "low-ohmic" for 3V VDDA – especially if the bridge resistance is high)
- BSS169: only 100V  $V_{DS}$  (6.8V Z-diode at generated VDDA recommended)
- BSS149: a bigger package
- BSP149: SOT-223-package (also bigger one)

### 7.2 External Clock Input Option

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### 8. RELATED DOCUMENTS

- ZMD31050 Feature Sheet
- ZMD31050 Datasheet
- ZMD31050 Application Kit Description
- ZMD31050 Development Status Report (including parts identification)
- ZMD31050 Calibration DLL Description
- ZMD31050 Application Notes

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#### For further information:

ZMD AG  
Grenzstrasse 28  
01109 Dresden, Germany  
Phone +49 (0)351-8822-366  
Fax +49 (0)351-8822-337  
[ssc@zmd.de](mailto:ssc@zmd.de)  
[www.zmd.de](http://www.zmd.de)

ZMD America, Inc.  
201 Old Country Road, Suite 204  
Melville, NY 11747, USA  
Phone +01 (631) 549-2666  
Fax +01 (631) 549-2882  
[sales@zmda.com](mailto:sales@zmda.com)  
[www.zmda.com](http://www.zmda.com)